



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013

Itgalpura, Rajankunte, Yelahanka, Bengaluru – 560064



BioRealm – Virtual Zoo & Educational Platform

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this report “BioRealm – virtual zoo & educational platform” is a Bonafide work of Deesha Mitra (20221CSE0064) , Swarnadeep Dutta (20221CSE0150)”, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE ENGINEERING during 2025-26

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE ENGINEERING at Presidency University, Bengaluru, named Deesha Mitra and Swarnadeep Dutta, hereby declare that the project work titled **“BioRealm – virtual zoo & educational platform”** has been independently carried out by us and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE ENGINEERING during the academic year of 2025-26. Further, the matter embodied in the project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

The preservation of animals and the education of the society about the richness of nature are two important issues that humanity faces today. As educational and recreational centers, traditional zoos have been around for a long time; nevertheless, they have problems such as the welfare of animals, access to the people living far away, and the safety of visitors physically. Moreover, the cost and effort of running real zoological parks are tremendous. This initiative offers "BioRealm," a deeply immersive virtual wildlife platform, as a digital answer to these problems.

BioRealm is an online service that aims at the recreation of a wildlife sanctuary spread worldwide. The model merges the sound and video libraries of ultra-resolution along with a 360-degree virtual safari element so that the users can visit the earth's different ecosystems without any restriction in the real world. In contrast to traditional learning methods, BioRealm takes an engaging method that is a quiz component with gamified features used for interaction and thus the learning gets reinforced which was based on animal behaviors, scientific classifications, and habitat conservation. This work made use of present-day internet technologies such as HTML, CSS, and JavaScript for the user interface and immersive media integration for virtual experience.

The creation of BioRealm is a testament to the idea that online zoos might be the right, easily expandable, and feasible alternatives to large scale physical captivity. To attract the users, as one of the main features, BioRealm presents a safe, easy, accessible and "Edutainment" style platform thus it can be considered as a practical solution to the issues stated by the United Nations Sustainable Development Goals (SDGs) in terms of quality education and life on land. U.N. SDGs propound quality education and life on land as two major issues which this digital platform named BioRealm takes a sizable leap in solving these problems providing an interesting platform of 'edutainment' in a safe, accessible, and engaging nature. This document comprehensively explains the research approach, the design, the execution, and the appraisal of the BioRealm team to the next generation of wildlife education.

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Abbreviations

Abbreviation	Expansion
2D	Two-Dimensional
3D	Three-Dimensional
4G	Fourth Generation (Mobile Network)
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
CC0	Creative Commons Zero (Public Domain)
CDN	Content Delivery Network
COPPA	Children's Online Privacy Protection Act
CPU	Central Processing Unit
CSS	Cascading Style Sheets
DOM	Document Object Model
FPS	Frames Per Second
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
GSAP	GreenSock Animation Platform
HTML	HyperText Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IDE	Integrated Development Environment
IoT	Internet of Things
IoTWF	Internet of Things World Forum
JPEG	Joint Photographic Experts Group
JS	JavaScript
JSON	JavaScript Object Notation
MDLC	Multimedia Development Life Cycle
MIT	Massachusetts Institute of Technology (License)

MP3	MPEG-1 Audio Layer III
MP4	MPEG-4 Part 14
PESTLE	Political, Economic, Social, Technological, Legal, and Environmental
PII	Personally Identifiable Information
PWA	Progressive Web Application
RAM	Random Access Memory
SDG	Sustainable Development Goal
SSD	Solid State Drive
UI	User Interface
URL	Uniform Resource Locator
UX	User Experience
VR	Virtual Reality
WebGL	Web Graphics Library
WebXR	Web Extended Reality

Chapter 1

Introduction

1.1 Background and Motivation

The Connections between human civilizations and wildlife is an inseparable part of the history where both parties have always been dependent on one another. To know more about the wildlife, traditional zoos were brought into picture. But is it really ethical to captivate animals just for showcasing them to humans? It is more than that, the deep relational between humans and wildlife should be a bond of respect rather than being a showpiece to entertain humans for fun. Traditional zoos are fun but not always accessible to everyone due to the fast-moving world and lack of time or lack of enough locations to visit a zoo. And even if traditional zoos are accessible, it ends up as a fun interaction and picnic rather than educational and informative. Keeping these things in mind BioRealm is an attempt to make zoos not only fun but also interactive and informative where one can study animals in depth and learn interesting facts about them. This project attempts to serve a sustainable and totally ethical way of learning about wildlife with the help of audio visuals and 3D tours without disturbing and disrespecting the wildlife thus spreading global awareness about animal rights and environmental ethics.

The motivation behind this project is that, education should go hand in hand with ethics and modern technologies rather than harming innocent wildlife and creating a disturbance in their environment. The philosophy is that technology can co-exist with nature at peace.

1.2 Statistics

The need for animal contact is great. Zoos and aquariums in the US receive more than 183 million visitors each year. This represents a deep human longing for an interaction with the natural world. But in-person interactions often lead to “compassion fade,” where the large number of animals or barrier of cages hampers a visitor’s ability to emotionally bond with wildlife.[5][10]

Additionally, two-thirds of adult zoo visitors attend with a child, underlining the educational role of zoos. But with hardship in travel expenses, seasonal weather, and global health crises (pandemics) over recent years has limited the physical access to

such learning centers. For a flight, we can't maintain the engagement statistics associated with an in-person event without the physical barriers.[5]

1.3 Prior existing technologies

Wildlife education in the past has mainly been through:

- i. Textbooks and 2D Images: These are informative but lack the interactivity which is very necessary when trying to engage the younger audience.
- ii. Physical Zoos: In these, observation is possible but limited to the animals' sleeping patterns, invisibility, and also the ethical issues of captivity.
- iii. Video Documentaries: These give visual context but they are passive experiences and hence the viewer has no control over their exploration.

One of the early virtual attempts was the "Virtual Reality Gorilla Exhibit" in the late 1990s that was directed at teaching students the social structures and vocalizations. After that, mobile apps with Google Cardboard have been trying to produce a stereoscopic view of animals. BioRealm goes further than these by supporting a very detailed 360-degree view that can be accessed on a browser and hence no need for any special hardware other than a computer or a smartphone.[8][7]

1.4 Proposed approach

1.4.1 Aim:

To develop a comprehensive web application "BioRealm" that serves as a Virtual wildlife experience and a wildlife encyclopedia with immersive safari.

1.4.2 Motivation:

The primary motivation behind this project is to make it to make zoos accessible by not putting the wildlife into unethical captivity and sufferings. By creating a virtual zoo, we can preserve the dignity of the animals while also making learning about wildlife fun and interactive.

1.4.3 Proposed approach:

The project uses a WebGL based architecture to portray the animals and their details. Few of the highlighted features are as follows:

- i. **Multimedia Library:** A vast collection of high-definition images and crisp clear audio recordings of animal calls have been incorporated.

ii. **Virtual Safari:** A 360-degree panoramic window has been incorporated which gives the user an experience of walking through the wildlife habitats, making them feel physically present at the location learning about the animals and their living conditions.

iii. **Interactive Assessment:** A quiz model has been incorporated to make learning about wildlife fun and informative. The quiz models emphasize on food chains which goes around in the habitats and interesting facts on wildlife. The quiz model dynamically gives the results to the users to boost their confidence.

iv. **Maps:** the regions a particular specie is to be found is displayed with all the information of the animals to know exactly at which regions a specie resides.

1.4.4 Applications:

There are a few applications of this project listed below:

i. **K-12 Education:** Could be used as an alternative to Biology(Mainly Zoology) classes regarding wildlife.

ii. **Remote Tourism:** Allowing less fortunate users or users with a medical condition/disability to experience wildlife habitats.

iii. **Conservation Awareness:** promoting ethical and empathic awareness towards endangered species.

1.4.5 Scope and Limitations:

1.4.5.1 Scope:

The scope encompasses the implementation and evaluation of a fully functioning web based virtual zoo prototype using HTML, GSAP ,Vanilla JavaScript and Pannellum.js. The most important and highlighted inclusions involve 4 models. 1) 360-degree virtual safari , 2) Ecosystem challenge quiz, 3) Audios of the habitat and species 4)explorer's map.

1.4.5.2 Limitations:

The current web application is a frontend only application which is working on a pre- loaded JSON database, lacking a real-time backend function. The virtual safari is created using panoramic images in 360-degree view using Pannellum.js instead of a 3D rendered worlds. The project currently relies on open-source for its information.

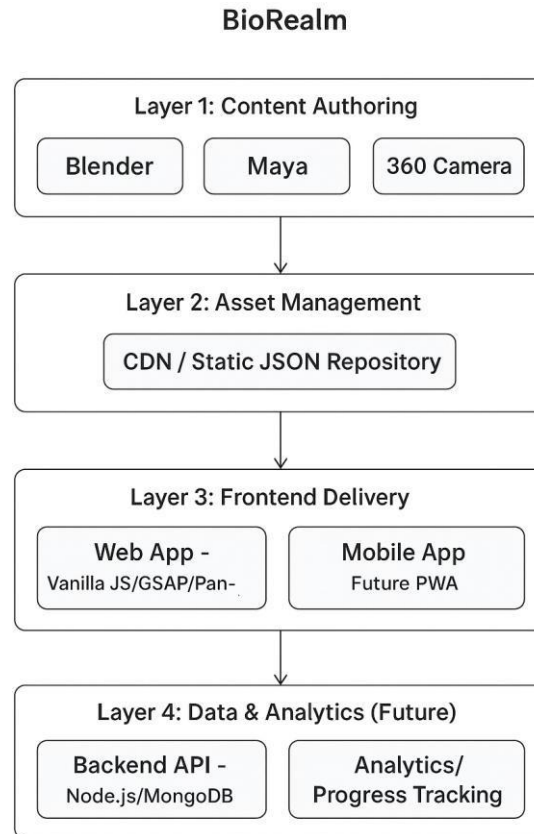


Fig. 1.1 : Conceptual Architecture of the BioRealm Platform

1.5 Objectives

The primary objective of this project is to create a virtual zoo atmosphere that promotes the awareness of biodiversity, conservation ethics and is also educational. The objectives include:

- i. Design an immersive user interface to stimulate a real jungle experience through organic and earthy theme and layouts.
- ii. To develop a high-performance web application using Vanilla JavaScript, HTML5 and CSS3 to ensure the cross-browser performance and speed.
- iii. Implement appealing and swift animations using GreenSock Animation Platform (GSAP) to enhance the virtual experience.
- iv. Create fun and engaging learning component such as an interactive quiz and 360-degree virtual safari.
- v. Ensuring that the web application interface isn't complicated for any groups of individuals to access.
- vi. Introducing realistic audios to make the user feel like they have entered a jungle safari.

- vii. Promoting ethical education and conservation awareness amongst the user to imply the sense of free will of the wildlife.
- viii. Build a scalable architecture for future integrations and expansion.
- ix. Incorporate a feedback mechanism such as a quiz to make the web application more interactive and engaging.
- x. Encourage research, collaboration and open innovation by making it open sourced and transparent.

1.6 SDGs

The United Nations Sustainable Development Goals (SDGs) provide a shared blueprint for peace and prosperity for people and the planet, now and into the future. The 17 goals, depicted in Figure 1.1, address global challenges including poverty, inequality, climate change, environmental degradation, peace and justice.



Figure 1.2: UN Sustainable Development Goals (Source: United Nations)

The BioRealm project aligns its core objectives with 3 SDGs, showcasing the commitment and dedication to the global sustainability. BioRealm specifically aligns with the following goals:

SDG 4 (Quality Education): BioRealm ensures learning opportunity for everyone by giving access to learning materials on wildlife and ecosystems. The interactive and engaging design promotes the interest of user to eagerly learning about the nature.

SDG 13 (Climate Action): the digitalized nature of the application talks about the climate impact. By having a virtual environment to study and learn about wildlife, we can observe a reduction in the carbon footprints that are associated with maintaining traditional zoos such as transporting animals from their habitat to a confined area and barely trying to recreate their original habitat.

SDG 15 (Life on Land): By promoting a virtual interactive model. BioRealm projects the need to protect the terrestrial ecosystem and spreads awareness amongst the users about conservation status, habitat destruction and exploitation of animals by migrating them to unfavorable location just for entertainment.

1.7 Overview of project report

This report is structured in a chronological order covering all the details on the BioRealm project. Chapter 1 gives an introduction on the concepts of BioRealm along with the objectives. Chapter 2 gives an analyzed report on existing literature papers written on virtual zoo and immersive learning. Chapter 3 gives detailed information on the methodology used in the development process. Chapter 4 outlines the project timeline along with the risk analysis and project budget. Chapter 5 shows details of the design and system analysis. This includes block diagrams, architecture and flowcharts. Chapter 7 shows the results and evaluation through testing of the project. Chapter 8 discusses the social, legal, ethical and sustainability aspects of the project. Finally, Chapter 9 summarizes the work and suggests future scopes.

Chapter 2

Literature review

Concepts such as virtual zoos are not at all new to us. In fact, over the last ten years; there has been a tremendous amount of work done by researchers, teachers, and technology developers on developing digital platforms to mimic, build upon or completely change the way we learn when it comes to zoos. With these digital platforms, the idea of creating virtual zoos was born due to many of the ethical issues surrounding the capture of animals, the financial burden of operating physical zoos, and the rising use of interactive digital technologies.

Virtual zoos can be designed to fit a wide range of formats, from simple static two-dimensional web pages displaying basic information about animals to advanced three dimensional, or VR based simulations allowing users to dynamically engage with digital wildlife.

2.1 Reviewed Research Papers:

2.1.1 Virtual Zoo: Immersive Digital Wildlife (Raichurkar et al., 2025):

This article will explore how to create a Virtual Zoo using Unreal Engine 5 for 3d modelling and PHP/MySQL for back-end data management. Authors of this study point out that digital technologies provide ways to develop environments that more accurately reflect the actual world. An important finding for BioRealm is their modular approach to system development. The authors divide the system architecture into a front-end user interface and a back-end database. This supports BioRealm's choice to divide the visual display of the quiz and the logic of the quiz data.[1][1][1]

2.1.2 VR Chiriyakhana: A Virtual Zoo Using Google Cardboard (Ahmed & Hossain, 2020):

Ahmed and Hossain developed an android-based app that simulated a virtual zoo. The authors found that combining natural sounds, with visual data provided a "raw feeling of nature", that increased both the user's relaxation and interaction with the application. They also noted that 360-degree video enabled users to view animals much more clearly than they could in traditional zoos. This supports BioRealm's inclusion of an

audio library, and 360-degree safari components to increase user immersion.[7][7]

2.1.3 Design of Virtual Reality Zoos Through IoT (Sukmawati et al., 2023):

This Research identifies the shortfalls of the physical zoo model including limited information on zoo signage and the limitation to being able to get close to animals as a result of safety issues. Authors developed a Virtual Reality Zoo using the Multimedia Development Life Cycle (MDLC), their results indicated that Virtual Environments are a viable educational medium for identifying wild animals safely. BioRealm adopted the MDLC model proposed in this paper as the developmental framework.[3][3]

2.1.4 A Case Study Exploring VR in the Zoo Context (Lugosi & Lee, 2021):

Lugosi and Lee studied the differences between learning outcomes from viewing videos versus experiencing VR regarding African Wild Dogs. While Lugosi and Lee found that the experience was viewed as “more enjoyable” and “naturally occurring,” some of the younger subjects experienced decreased retention of factual information due to distraction caused by the visual elements. This is an important factor for BioRealm as to address this issue of “distractions,” BioRealm has included a Quiz Component so that users have to recall and integrate what they have observed to ensure that the platform will be an educational resource rather than simply entertainment.[8][8]

2.1.5 The Effects of Embodying Wildlife... (Pimentel & Kalyanaraman, 2022):

This study examined “Compassion Fade,” which is the decrease in concern over mass casualty versus individual victims.

Authors demonstrated that through the act of virtually embodying a species in peril (i.e., sea turtles), individuals are more likely to exhibit an intent to donate and contribute to conservation efforts when using a virtual environment. This is supportive of BioRealm's goal of promoting empathy and compassion. Through the use of 360-degree views of the habitat within BioRealm, the developers hope to elicit an embodied cognition response, thereby promoting conservation awareness.[10][10]

2.1.6 Zoo Therapy for Alzheimer's Disease (Ai et al., 2021):

Although the focus of this paper was therapy related, it illustrated that users may benefit cognitively and emotionally from interacting with virtual animals (e.g., horses, dogs). The system used 3D buttons and voice commands for interaction. The results validated the potential for BioRealm to serve as both an educational and potentially a therapeutic resource to promote relaxation based upon the calming aspects of nature sounds and visuals of animals.[6][6]

2.1.7 Unveiling Zoological Realms (Ugli et al., 2024):

The literature review provided a comparative analysis of the exportation of three-dimensional models for virtual zoos with an emphasis on OBJ, DAE, and FBX formats; as such, it was determined that while FBX files are the most compact, they have scalability limitations. The literature also informed the selection of lightweight media format(s), including glTF and/or optimized MP4 360-degree video, to enable rapid loading of the application in a web-based browser environment for BioRealm.[2][2]

2.2 Overview of the papers

2.2.1 Existing Virtual Zoo Platforms and Limitations

Some of the first attempts at creating virtual zoos used a variety of game engines including but not limited to Unity 3D and Unreal Engine. Both of those engines can create very realistic 3D models and have real-time physics that allows them to model the environment and animals in a way that mimics reality, with lifelike animation. However, both of those engines have had limitations based upon their need for extensive amounts of computational power and specialized hardware (i.e., VR Headset or High-End Graphics Cards) to run smoothly, which has limited their reach to only a small number of users. Additionally, many of the applications developed using these engines suffered from a "WebGL Bottleneck" when they were attempted to be loaded into web browsers, especially on mobile devices; causing the frames to drop, the loading time to be excessively long, and also overheated the device. The balance between immersion and accessibility are the two major technological limitations.

2.2.2 Pedagogical Theories in Virtual Environments

To effectively support education in virtual environments they must support learning in

all three areas of learning: cognitive, emotional, and experiential. Based on Constructivist Learning Theory, where students develop their own knowledge through interaction and experience rather than through being passively shown information; the development of BioRealm was guided by this theory. In a virtual environment, students are able to interact with the environment in real-time and receive dynamic feedback and can explore the environment and receive manipulation of objects and receive feedback and therefore tend to gain deeper conceptual understandings of subject matter, such as zoology and ecology. Another approach that has been successful in supporting student learning and engagement in a virtual environment is Gamification. By applying game design elements (such as challenges and rewards), it creates an active form of exploration and motivates students to learn, while also retaining the knowledge learned during the process of observing.

2.2.3 Technical Implementations and Web Technologies

Educational Systems have combined different multimedia elements – audio, video, and interactive – in one field. Previous research describes the development of 3D content utilizing pipelines such as Blender → FBX → Unity. To optimize these resources to be used effectively in an educational setting; previous research indicated that the use of optimization methods were required (for example mesh decimation, LOD scaling). A difference in this approach is the focus on a light weight browser based structure. The use of Vanilla JavaScript with GSAP and Pannellum.js is intended to provide the same level of technical consistency across all client devices as possible while also maintaining speed and performance and eliminating the need for large framework usage (such as React or Angular) to ensure maximum accessibility.

2.2.4 Gamification in learning

The application of gamification—utilizing game design principles in non-game related settings—has been shown to have an impact on motivation, engagement and knowledge retention. Al-Araibi et al. (2021) developed a conceptual model linking gamification and sustainable learning behaviors. Gamification elements such as point systems, reward structures and interactive challenges can enhance the learner's intrinsic motivation.[11]

Virtual zoos are another setting where gamification takes place by changing observational experiences into exploratory ones. Virtual zoo visitors are able to engage

in interactive games/challenges i.e., build food chains, identify different animal species, or complete adaptive quizzes. The "Ecosystem Challenge" within BioRealm utilizes many of the above-mentioned techniques by allowing learners to demonstrate their knowledge in a visual/cognitive manner. The learner receives visual and cognitive reinforcement for each successful response through the use of animated cues; this creates an experience of accomplishment that supports learning.

Table 2.1 Summary of Literature reviews

Sl.	Article Title / Author/Year	Methods	Key Features	Merits	Relevance to BioRealm
1	Virtual Zoo: Immersive Digital Wildlife (Raichurkar et al., 2025)	Unreal Engine 5, PHP, MySQL	Realistic 3D biomes, Web-based interface	Accessible via web, high graphical fidelity	Validates web-based architecture for accessibility.
2	VR Chiriyakha na (Ahmed & Hossain, 2020)	Unity 3D, Google Cardboard	360-degree video, Natural sound integration	Low cost, immersive audio-visuals	Supports the use of 360 visuals and audio for immersion.
3	Design of VR Zoos Through IoT (Sukmawati et al., 2023)	MDLC (Multimedia Development Life Cycle)	Mobile VR for animal recognition	Safe interaction with dangerous animals	Provides the MDLC methodology used in BioRealm.
4	VR in the Zoo Context (Lugosi & Lee, 2021)	Semi-structured interviews	Comparison of Video vs. VR learning	VR is more enjoyable and naturalistic	Highlights need for Quizzes to ensure fact retention.
5	Embodying Wildlife (Pimentel & Kalyanaraman, 2022)	Virtual Reality Experiments	User embodied animal avatars	Increases empathy and conservation intent	Supports BioRealm's goal of conservation awareness.

2.3 Research Gaps of Existing Methods

The current state of virtual wildlife education has some significant gaps in the literature review for which BioRealm seeks to find answers:

- **Performance vs. Accessibility Bottleneck:** High-quality VR/3D systems are

often limited by access to the necessary equipment to run such systems; the gap exists as there currently is no high-performance, low-cost, immersion system.

- **Lack of Pedagogical Depth:** Current technology focuses mainly on creating visually appealing rendering experiences (for example: high quality visuals), while the educational aspects have been left out including an evaluation plan, and clear educational pathways.
- **Passive Engagement Models:** Most of the systems currently available use pre-recorded 360° videos or still models that users can view passively, and do not allow for the active exploration and discovery process that is needed to achieve retention of information.
- **Technological Unsustainability:** Relying too heavily on large frameworks (such as React, Unreal, Unity) will create large, complex systems that are difficult to maintain and require a lot of resources to operate, which contradicts many of the long-term conservation goals of this project.

Table 2.2 Summary of Identified Research Gaps and BioRealm’s Solutions

Research Gap	Observed Issues in Existing Systems	BioRealm’s Approach / Solution
Performance & Accessibility	Heavy 3D rendering and VR requirements limit accessibility. WebGL lag affects mobile devices.	Lightweight, browser-based implementation using Vanilla JS, GSAP, and Pannellum.js ensuring fast load times and universal compatibility.
Lack of Pedagogical Integration	Systems focus on visuals, neglecting structured educational design or assessments.	Built-in gamified quizzes, guided learning modules, and a knowledge evaluation framework.
Passive Learning Models	Users act as spectators rather than participants.	Interactive, discovery-based learning through drag-and-drop, exploration, and immersive challenges.
Technological Unsustainability	Framework-heavy architectures consume more energy and are complex to maintain.	Minimalist, static-first design that prioritizes sustainability and scalability.

Chapter 3

Methodology

In order to better meet the needs of a bio-realistic, interactive, virtual reality (VR) experience; the BioRealm Project implemented an Agile method as a base and modified it to fit the specific parameters of a small group, capstone project which was time-bound and required a significant amount of iteration and integration of both design and technical components. The method also provided an environment where developers could rapidly prototype and adapt, allowing them to combine rigorous software development with subjectively driven design principles such as aesthetic appeal and user experience.

This process is conducted in short, iterative loops, with each loop culminating in the delivery of a minimum viable product module prior to proceeding to the next. This ensured early bug detection and immediate incorporation of user feedback regarding immersion and usability, which are core objectives.

3.1 Methodology Selection

Sukmawati, et al. state that the Multimedia Development Life Cycle (MDLC), chosen as the primary methodology for developing BioRealm, is especially suited to projects that combine large amounts of media resources (graphics, audio, video) with software programming. The MDLC approach differs from the Waterfall method in its linear nature; and from the Agile methodology where all focus is placed on iterative changes to the coding, in that it emphasizes the creation and combination of media resources — the heart of BioRealm.[3]

3.2 Stages of MDLC in BioRealm

3.2.1 Concept:

The Concept phase represents the initial stages of the project, where the goal(s) and the scope of the project are determined.[3]

- **Goal :** The Virtual Zoo aims to provide a simple, web-based environment in which to virtually explore wildlife habitats with no need for actual travel to such locations.
- **Primary Target Audience :** The main target audience consists of two groups. These include K-12 and college-level students who require educational information

about wildlife and other wildlife enthusiast interested in a highly immersive experience.

Three core features are envisioned for the application:

1. An informational database.
2. A 360-degree virtual safari module.
3. A quiz module to assess the user's knowledge.

3.2.2 Design

In this design phase, the specifics for the media and architecture of the Virtual Zoo application were determined.[3]

User Interface (UI): Wireframes were created to illustrate the user experience throughout the site. This included the Home page, navigation path through the different "Biomes," and the User Interface for the Quiz component.

Storyboard : A Storyboard was developed to outline the order of events in the 360-degree virtual tour so that the user will be able to easily navigate from a large panoramic view to individual animals.

Technical Architecture: It was decided to utilize a client-side architecture with HTML5, CSS3 and JavaScript to allow for cross-platform compatibility for use on all desktop and mobile devices.

3.2.3 Material Collection

At this stage the developers gathered together the original components that would be used to build the application.[3]

- i. **Visual Resources:** High-resolution images of animals and equirectangular (360 degree) panoramic images of several natural environments were acquired through free access image resources and licensing libraries.
- ii. **Sound Resources:** Authentic sound recordings of animal vocalizations (e.g. Lion Roar, Elephant Trumpet) and sound effects of the environment (Wind, Water Flow) were used to further enhance the naturalistic experience of the virtual safari simulation.
- iii. **Written Content:** Educational information concerning the scientific classification (binomial nomenclature), diet, and the current conservation status of the animals featured in the simulation were compiled from reliable zoological data sources.

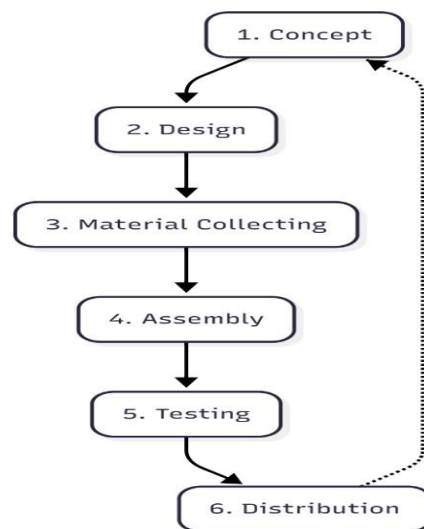


Fig. 3.1 MDLC life Cycle

3.2.4 Assembly

During this phase the developers incorporated the material that had been previously gathered into a singular, working application.[3]

- i. **Development:** The application was developed with Visual Studio Code. HTML was utilized to define the content structure, while CSS was used to create a flexible and responsive design that could adapt to the different dimensions of screens.
- ii. **Assembly/Integration:** JavaScript Libraries (Pannellum or A-Frame) were assembled into the application so that 360-degree images could be rendered, providing users with the ability to manipulate their viewpoint by dragging the mouse or touching the screen.
- iii. **Implementation of Logic:** JavaScript code was created to implement the quiz logic; this included tracking user's responses, calculating scores in real time and giving instant feedback on whether a response was correct or incorrect.

3.2.5 Testing

Prior to the final release of the application, testing was implemented to verify that it functioned as desired.[3]

- i. **Functional Testing:** Each link, button, and media player were tested to confirm that they reacted properly to user interaction.
- ii. **Compatibility Testing:** To confirm that the 360-degree rendering engine was consistent across multiple web browsers (Google Chrome, Mozilla Firefox,

Microsoft Edge), compatibility testing was performed.

iii. **Alpha Testing:** A small number of users experienced the "Virtual Safari" to determine if there were any lags or inconsistencies between the audio and visual elements.

3.2.6. Deployment

Making the application available to the end-user for use was the last phase in this process. The BioRealm project was deployed onto a general public web hosting service (i.e., GitHub Pages/Vercel).[3]

End User Access: The final product was organized into a file structure so that all large media can be loaded quickly and efficiently by the user. The final product allowed users to access the Virtual Zoo using a simple URL link, thus meeting the objective of global access.

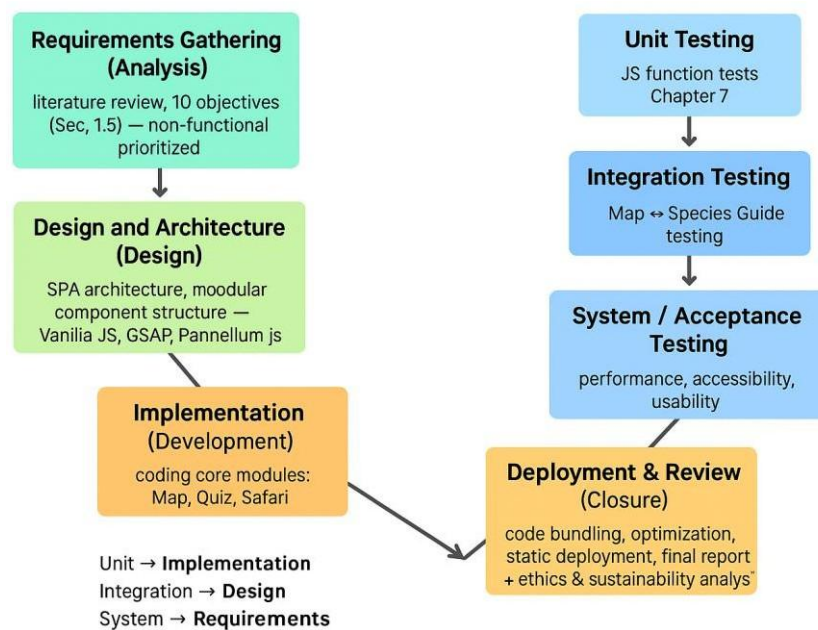


Fig. 3.2: V-Model (methodology) of BioRealm Platform

Chapter 4

Project Management

4.1 Project timeline

The Project Timeline is an overview of all the work activities involved in developing BioRealm. The project was broken down into two separate, but related stages to create order and process to the project; Phase I: Project Planning and Design, and Phase II: Project Development and Testing.

By doing so we could focus on building the theoretical framework and design requirements before we actually begin writing code or integrating media into our project.

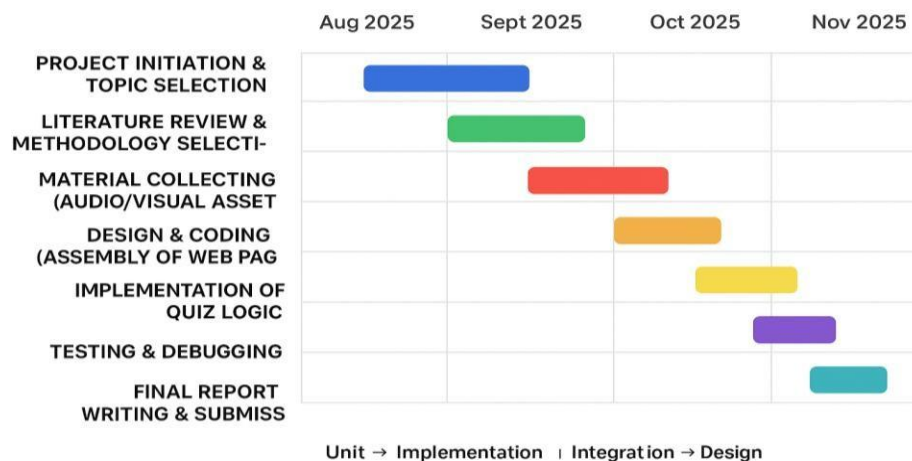


Fig. 4.1 Project Timeline

4.1.1 Phase I : Project planning Timeline

The Planning Phase addressed the development of the Blueprint for BioRealm; defined the boundaries of the BioRealm problem space through literature surveys; and developed an Architectural Framework for the Web Application. These initial phases took place during the first 8 weeks of the project.

Weeks 1-2 (Initiation and Background): In the beginning of the project, we identified the central issue that BioRealm would be addressing: Zoos are physically inaccessible to many people and there are growing concerns about animal welfare and the ethics of keeping animals' captive. We established the topic of "BioRealm" and decided to apply digital solutions to both problems. At this point, the primary goals for the project were formally articulated: creating a web-based, educational, and immersive experience for users.

Weeks 3-4 (Literature Review): A comprehensive review of relevant literature was performed. We reviewed papers such as Virtual Zoo: Immersive Digital Wildlife and VR Chiriyakhana to understand the current state of virtual reality within the area of wildlife education. Through our research efforts, we determined that there was a significant gap between existing VR applications and a web-accessible (not requiring headsets), thus influenced our decision to utilize WebGL/JavaScript over heavy gaming engines like Unreal Engine.[1][7]

Week 5 (Methodology): The Multimedia Development Lifecycle Model (MDLC) was selected as the overarching model to guide our methodology due to the dependency of our product on multimedia. Additionally, we utilized the V-Model for the software engineering portion to ensure adequate and thorough testing.[3]

Weeks 6-7 (System Design): During this critical phase, we designed both the visual and logical organization of the application. We produced wireframes for the Home Page, Safari page and Quiz page. We also created a System Flow Diagram to illustrate the logical sequence of the User Experience from Safari (observation) to Quiz (assessment).

Week 8 (Material Collection): Due to the content-intensive nature of BioRealm, we dedicated a considerable amount of time to locate high definition Equirectangular (360 degree) Images and Authentic Animal Sounds. We had to verify the copyright license (Creative Commons) to ensure we were compliant with all applicable laws.

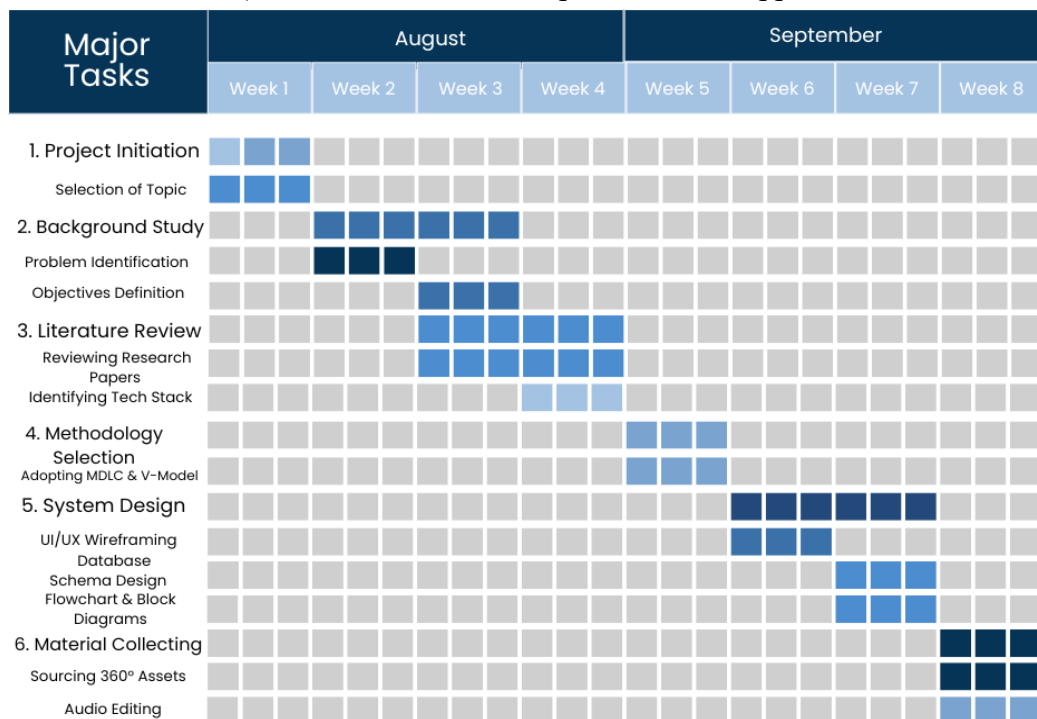


Fig. 4.2: Project Planning Gantt Chart

4.1.2 Phase II: Project Implementation Timeline

The Implementation Phase is where you would actually write the code and deploy your Web Application. It also includes the testing of your application. This phase requires a great deal of technical detail so that the 360-degree rendering engine works well with all types of devices. Weeks 9 through 15 are part of this phase.

Weeks 9 & 10 (Front-end Development): The base of your application is created in HTML5. Your application's look is defined by CSS3. In addition, you will define a visual theme that simulates a movie theatre experience when viewing the pictures.

One of the biggest focuses of this development stage is making sure that your application is responsive. So, it can be viewed properly on both desktop and mobile screens.

Weeks 10 & 11 (Logic Creation): This is the point at which you create the JavaScript code that integrates the Pannellum (API) of the 360-degree viewer. This means you need to write the logic to map the 2D equirectangular images onto a 3D sphere and allow users to interact with them. Next, you will develop the logic for the Quiz; this involves creating arrays to hold questions, and functions to calculate and display the score as they progress.

Week 12 (Asset Embedding): You will embed the collected assets into the code. You will link audio files to buttons and optimize the high-resolution panoramas for fast web loading.

Week 13 (Testing): The testing process is very thorough. You will perform unit testing to make sure each module works separately. Then you will test if moving from the "Safari" page to the "Quiz" page preserves the user's session information. Also, you will conduct cross-browser testing on Chrome, Edge and Firefox to make sure the 360-degree rendering engine works correctly on all platforms.

Week 14 (Launch): After you have completed testing, the final version of your application will be launched to a live web server. This includes configuring your repository to make your website publicly available through a URL.

Weeks 14 & 15 (Documentation): The final project report will be written during this time. The report should include a description of every aspect of the development process, an analysis of the outcome of the project and be formatted according to your university's guidelines.

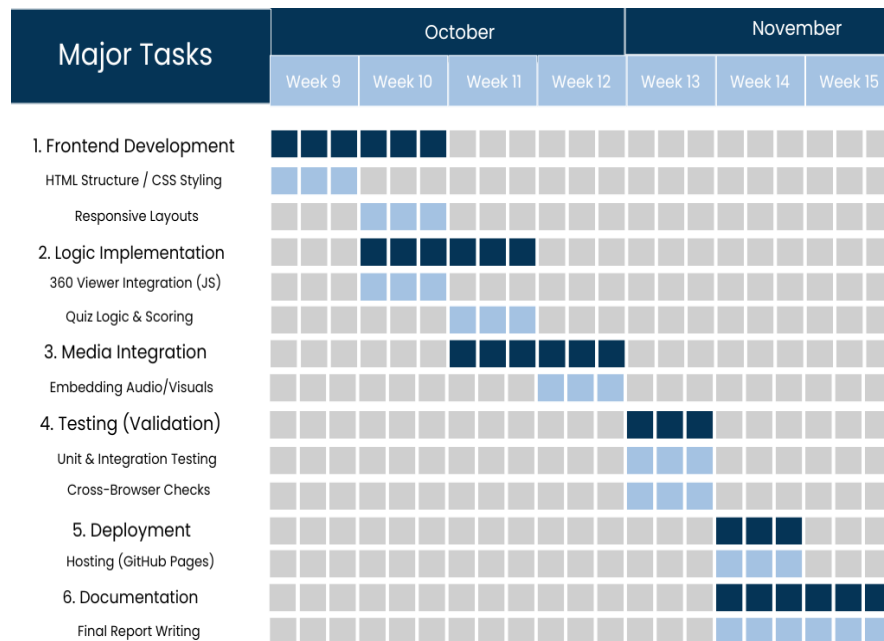


Fig. 4.3: Project Implementation Gantt Chart

4.2 Risk analysis

A major part of Project Management is Risk Analysis. Risk Analysis identifies possible threats which could jeopardize the success of the Project, as well as developing contingency plans in order to avoid those threats. In the case of BioRealm, Risks were broken down into two categories: Macro-Environmental (using the PESTLE Model), and Internal Execution of the Project (using a Probability-Impact Matrix).

Table 4.1 PESTEL analysis framework



4.2.1 PESTLE Analysis

The PESTLE framework examines the Political, Economic, Social, Technological, Legal, and Environmental factors that might impact the deployment of the BioRealm application.

Table 4.2 PESTEL Analysis for BioRealm

Category	Potential Risk Factor	Impact Description	Mitigation Strategy
Political	Educational Policy Changes	The following factors could limit users' ability to use this service due to restrictive Digital Education policies or Internet Censorship in specific countries	If internet access is limited, develop the service as a downloadable static website so it can be run offline.
Economic	Hosting & Asset Costs	High quality 360-degree content and the cost of cloud services are expensive.[1]	Use Open-Source media libraries (NASA, Unsplash), and Free Tier Hosting Options (GitHub Pages, Vercel), to make sure your project has a \$0 budget.
Social	Digital Divide	Students in rural areas with poor internet connectivity or older devices may struggle to access the high-fidelity 360-degree safari.	Implement "Low Bandwidth Mode" where 360-degree visuals are replaced with high-quality static 2D images for
			accessibility on slower networks.
Technological	Device Incompatibility	It is possible that older browsers or lower end smart phones do not support WebGL rendering engines needed for 360 views of animals.[1]	Conduct Cross Browser Testing (Chrome, Edge, Safari), and Develop Code Logic for Degradation if a browser does not support WebGL rendering engines.
Environmental	Carbon Footprint of Servers	Although Virtual Zoos Reduce Travel Emissions, Cloud Servers Consuming Energy Still Exist.	Reduce Data Transfer Sizes by Compressing Media Files, Therefore Reducing the Amount of Energy Required for Server Requests.
Legal	Copyright Infringement	Legal Action Could Be Taken If You Use Wildlife Images or Audio Without a Proper License	Only use CC0 or Public Domain licensed Media. Keep a Citation Log for all Media Used.

4.2.2 Project Phase Risk Matrix (Internal Risks)

In addition to using the PESTLE framework to identify external risk factors, the Project Phase Risk Matrix identifies internal technical and management risks that could arise during the process of developing BioRealm, and evaluates these risks according to their likelihood (probability of occurrence) and severity (project impact).

Table 4.3 Project Execution Risk Matrix

Risk ID	Risk Description	Likelihood (1-5)	Severity (1-5)	Risk Level	Mitigation Plan
R1	High Latency (Lag): The 360- degree panoramic images are large files. They might fail to load or stutter on mobile devices, ruining the immersion.[1]	4	5	Critical	Use image compression algorithms to reduce file size without losing visual quality. Implement "lazy loading" so images only load when needed.
R2	Scope Creep: Adding too many features might	3	4	High	Strictly follow the MDLC methodology. Stop adding more requirements after the
	exceed the time limit of the semester.				"Design" phase and move them to "Future Work."
R3	Browser Compatibility: The JavaScript library used for the 360- viewer might not work on Safari (iOS) due to security protocols.	3	5	High	Test on an iPhone in the early development cycle. If the library fails, switch to a more compatible alternative if available.
R4	Data Loss: Loss of code or assets due to hardware failure or accidental deletion.	2	5	High	Use Git for version control. Push code to a remote repository (GitHub) daily to ensure a cloud backup exists.

R5	Asset Unavailability: Not able to find good images and audio recordings	3	3	Medium	Flexible in animal listing inside the database. Shift to a looped video in such cases.
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4.2.3 Risk Conclusion

Technological Risks (R1, R3), are identified as the greatest risk to the project because of the project's heavy dependence on WebGL for rendering the 360-degree safari. The risk mitigation plan is to have the development team focus on performance optimization and testing on lower end devices during the implementation phase as suggested by Raichurkar et al. (2025). By mitigating these risks, the project expects to be able to provide an educational platform that is stable, accessible, and legal.[1]

4.3 Project budget

Project budgeting is an essential component in managing projects which ensures that a project will be successfully completed at a cost no greater than what has been allocated as budgeted funds. The BioRealm project used a "Zero-Cost" Budgeting Strategy exclusively. A Zero-Cost Budgeting Strategy involves using free (open- source) software and utilizing Free-Tier Cloud Services; thereby eliminating all financial expenditures associated with software licensing and/or cloud service fees without sacrificing either Quality or Functionality.

The following is a budget analysis comparing the estimated market cost of software license and cloud services costs to the actual zero-cost strategy employed by the team over the 14-week development period.

Table 4.4 Project Budget Estimation

S.No.	Particulars	Resource Type	Budgeted Cost (₹)	Actual Cost (₹)	Variance (₹)	Remarks
1. Software Development Tools						
1.1	Visual Studio Code (IDE)	Software License	0	0	0	Open Source (MIT License).
1.2	Blender (For 3D Editing)	Software License	0	0	0	Open Source (GPL License).
2. Graphics & Design Tools						

2.1	Adobe Photoshop / Canva Pro	Graphics Tool	1,500	0	-1,500	Used free alternatives
3. Cloud & Hosting Services						
3.1	Domain Name Registration (.com)	Service	850	0	-850	Used GitHub Pages subdomain (free).
3.2	Web Hosting Server	Service	2,500	0	-2,500	Deployed on GitHub Pages Free Tier.
TOTAL	Grand Total		₹ 4,850	₹ 0	- ₹ 4,850	Effective Cost

4.3.1 Budget Analysis

In order to make the project affordable, the project developers chose to use open-source development technology only. Due to the requirement to utilize premium graphics editing tools to edit 360-degree images in high resolution (approximately ₹1500) – the project developers used free alternatives like GIMP and the free version of Canva instead – therefore the actual cost of software for the project was approximately ₹0.

The estimated cost of cloud and hosting for the professional website that would be available globally was ₹3350; however, due to the use of GitHub Pages as the website's host – where there is no charge for static hosting or for a secure subdomain for an open-source project — the cost of cloud and hosting was completely avoided.

In summary, the estimated market value of the resources and services needed for the software was approximately ₹4850. As a result of efficient utilization of available resources, and utilizing open-source alternatives, the project was completed at a cost of ₹0, thus achieving maximum efficiency.

Chapter 5

Analysis and Design

The two key components that connect the conceptual aspect of BioRealm with its operational reality, are analysis and design; analysis is the process of identifying the educational and immersive aspects of the virtual zoo, whereas design identifies the architecture of the system as a whole and how it will be structured, and includes the logical connections of the data flow and user interface.

5.1 Requirements

BioRealm's system development will be based on these requirements which are both functional and non-functional in nature and have been developed through the literature review by considering the necessity for reproducing environments realistically and through the incorporation of interactive features to promote knowledge retention.[1][8][8]

5.1.1 Functional Requirements

These are the things that the BioRealm system has to do:

Rendering of Virtual Safari: The system must generate high resolution equirectangular images (360-degree panoramic views) of the virtual safari to allow users to rotate their view 360 degrees horizontally and 180 degrees vertically using mouse or touch controls.[7]

Playback of Audio Files: The system must obtain and playback audio files (animal calls, nature ambiance etc.) when the user interacts with the application.[7]

Quiz Logic: The system must prompt the user with multiple choice quizzes, receive input from the user, validate the user's answer against a stored value for the correct answer, and calculate an instantaneous score.[8]

Navigation: The system must provide the user with a seamless transition between the "Home", "Safari" and "Quiz" modules of the application without reloading the current page (Single Page Application behavior).

5.1.2 Non-Functional Requirements

These are the ways that the system should behave:

Latency: Media asset loading time of less than three seconds over a standard 4G

network is required to keep the user engaged.[2]

Responsiveness: The UI must dynamically resize and reorganize itself to fit screen resolutions for desktop (1920 x 1080), tablet, and mobile devices.

Compatibility: The rendering engine must support all of the major browsers (Chrome, Firefox, Safari, Edge) and operate without relying on external plugins or VR headsets.[7]

Scalability: The file structure of the system must be able to accommodate additional animal exhibits without modifying the core codebase.

5.2 Block diagram

To give you a good idea of what the overall system will look like, the block diagram depicts the key elements that make up the system and how the data flows between them. In the case of BioRealm, the system has been divided into the User Interaction Layer, the Processing Logic and the Data Storage.

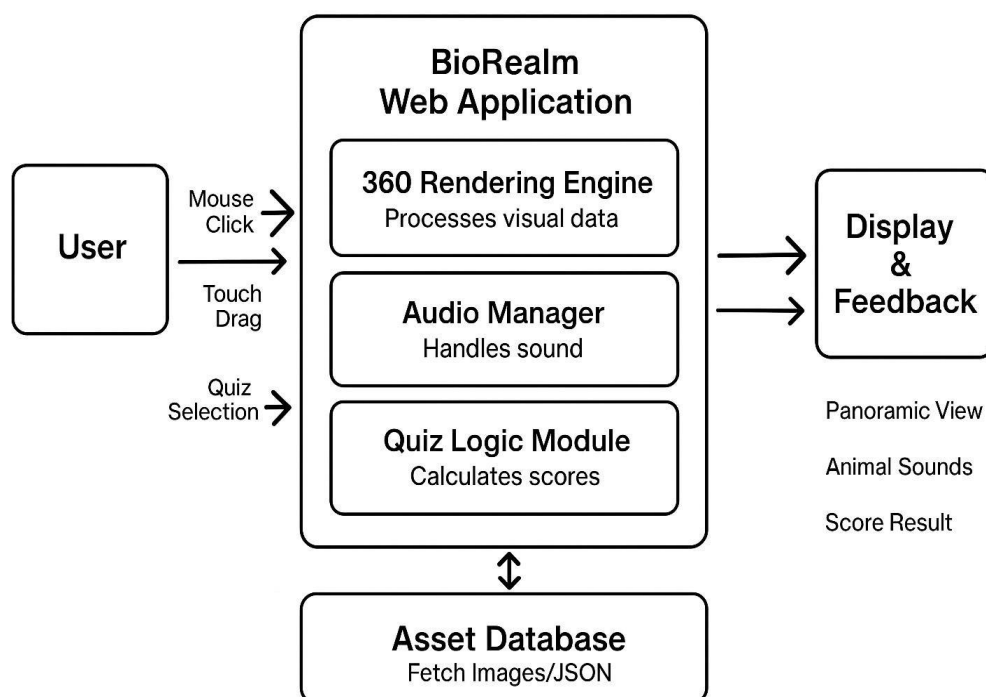


Fig 5.1 Functional block diagram of BioRealm System

5.3 System Flow chart

The System Flow Chart shows the users' progression through the application, outlining the choices and logical branching from the time the user enters the website to when they finish the educational assessment.

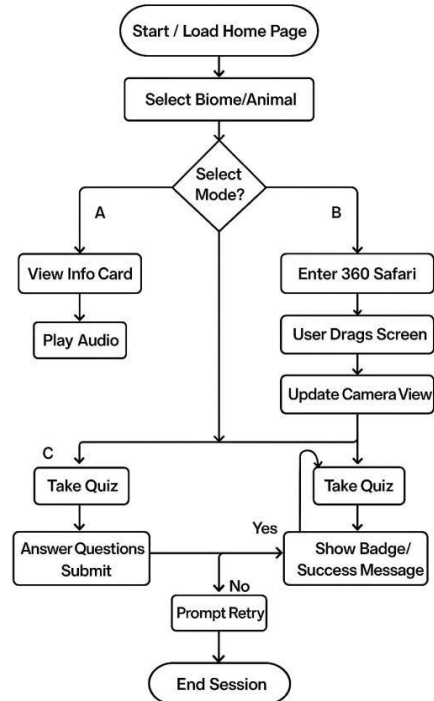


Figure 5.2: System Flow Chart for User Interaction

5.4 Choosing devices and Tools

BioRealm is a web application so "devices" refers to the development stack and the client-side requirements. Choosing the appropriate file format was a key decision since, as stated by Ugli et al., selecting the best format (such as OBJ or FBX for 3D objects, or an equirectangular JPG for 360-degree web views) is important to balance visual quality with file size.[2]

Table 5.1: Comparison of 360 Degree web viewers

Feature / Tool	Pannellum	A-Frame (WebVR)	Three.js	Selection for BioRealm
Ease of Use	High (Simple API)	Medium (HTML based)	Low (Complex Code)	Pannellum
File Size	Lightweight (15kb)	Heavy (1MB+)	Heavy (500kb+)	Pannellum
Performance	Excellent for static 360 images	Good for 3D models	Excellent but resource-heavy	Pannellum
Device Support	All Browsers	Requires WebXR support	All Browsers	Pannellum

Justification: Pannellum was used in place of other panoramic viewing software since BioRealm focuses on photographic 360-degree tours, not 3D polygon models.

Additionally, it has low overhead and thus does not expose the "latency" issues we found in our risk analysis.

Table 5.2: Comparison of Media Formats

Feature / Format	PNG	JPEG	WebP	Selection for BioRealm
Compression	Lossless (Large Size)	Lossy (Small Size)	Superior Compression	JPEG / WebP
360 Metadata	Poor Support	Good Support	Good Support	JPEG
Transparency	Yes	No	Yes	PNG (for UI Icons)

Justification: As indicated by the results of Ugli et al.'s research study, minimizing file size is a very significant consideration when working with images on the web. We chose to use JPEG for the 360-panoramas as they offer the highest compression ratio and can therefore be loaded quickly over the web.[2]

5.5 Designing units

To make development and testing simpler, the BioRealm application was divided into three functional areas.

1. **The Visualization Unit (The Safari):** This unit rendered the virtual environment and took a 2D equirectangular image as input to map the image onto a 3-dimensional space (a sphere). The users' viewpoint served as a camera located at (0,0,0) in the center of the sphere.

Input: Design Image 4096 x 2048 pixels.

Output: Interactive Canvas Element.

2. **The Assessment Unit (The Quiz):** This unit handled the educational logic and utilized a JavaScript array of objects to store each question's possible answers and the index of the correct answer.

Logic: If (User Choice = Correct Answer) {Score = Score + 1}.

3. The Interface Unit (UI/UX): This unit managed the layout and used CSS Grid to position the "Animal Info Card" next to the "360 Viewport" on desktop screens but stacked them vertically on mobile devices (responsive design).

5.6 Standards

Following established web and media standards helps BioRealm remain both secure and available to users.

Semantic HTML5 Standards: As part of our commitment to making sure BioRealm is accessible to all students, including those who are disabled and use screen-readers; BioRealm uses semantic HTML5 standards (including div, span, p tags etc.) to maintain compliance with SDG 4 (Quality Education).

Spherical Image Standards: All 360-degree images used in BioRealm have been created using an equirectangular projection which has a 2:1 Aspect Ratio. This ensures they can be properly mapped to a sphere without any distortion.

JavaScript Code Standards: Our JavaScript code follows ES6 (ECMAScript 2015) standards which allows us to utilize new syntax such as 'let/const' and 'Arrow Functions' to write clean, efficient and easy-to-maintain logic.

5.7 Mapping with IoTWF reference model layers

While BioRealm is a web application, it may also be viewed through the lens of the Internet of Things (IoT) World Forum Reference Model to illustrate how the data handling in BioRealm is similar to many modern virtual zoo applications.[1]

Table 5.3: Mapping BioRealm to IoTWF Layers

Layer	IoTWF Model	BioRealm Implementation
7	Application	The interactive Website (UI) where users engage with the safari and quiz.
6	Application Support	The JavaScript Logic (Quiz scoring, 360 rendering loop).
5	Data Abstraction	The JSON files storing animal facts and quiz questions.
4	Data Accumulation	The local browser cache (storing images for faster reload).
3	Edge Computing	Client-side processing (The user's laptop/phone CPU renders the graphics).

2	Connectivity	HTTP/HTTPS protocols transferring data from the server to the client.
1	Physical Devices	The end-user's Smartphone or Laptop.

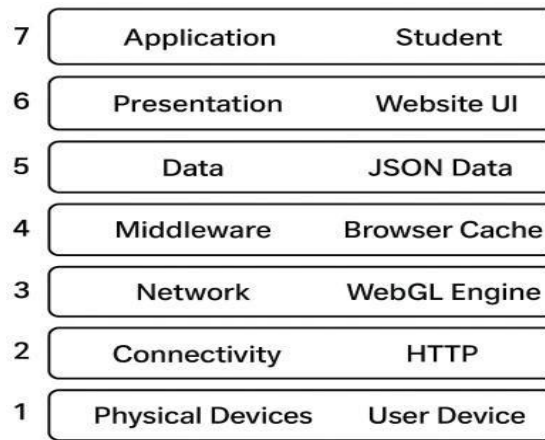


Fig. 5.3 IoT world Forum Reference Model

5.8 Domain model specification

The Domain Model illustrates the different entity types in the BioRealm system and how they relate to each other. While the Domain Model is abstracted from the specifics of the implementation, it clearly defines the overall structure of the data.

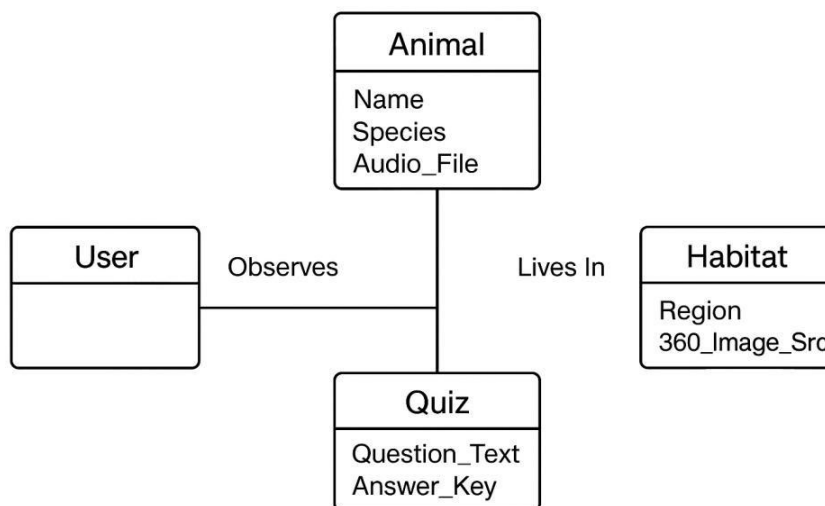


Fig 5.4 Domain model for BioRealm

5.9 Communication model

BioRealm utilizes a request-response communication model. In BioRealm, requests are generated by the client (the User's Browser) and responses are returned from the server (GitHub Pages).

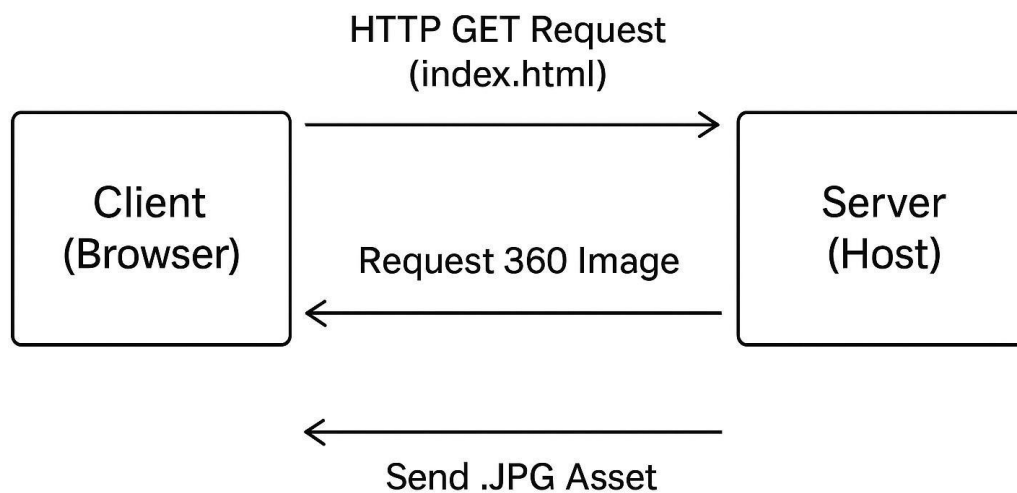


Fig 5.5 Communication model for BioRealm

5.10 IoT deployment level

BioRealm utilizes a level 2 deployment model.

Description: In a level 2 deployment model, the application logic and the visualization engine are executed locally on the client (edge or user device), and the data (code and images) is stored remotely in the cloud.

Suitability: A level 2 deployment model is particularly suitable for BioRealm, as the rendering of 360-degree graphics requires an immediate response to mouse movement. If the logic were executed on the server there would be noticeable delay. To avoid this delay, the code is downloaded to the client (level 2) and executed locally ensuring a seamless and lag free safari experience.

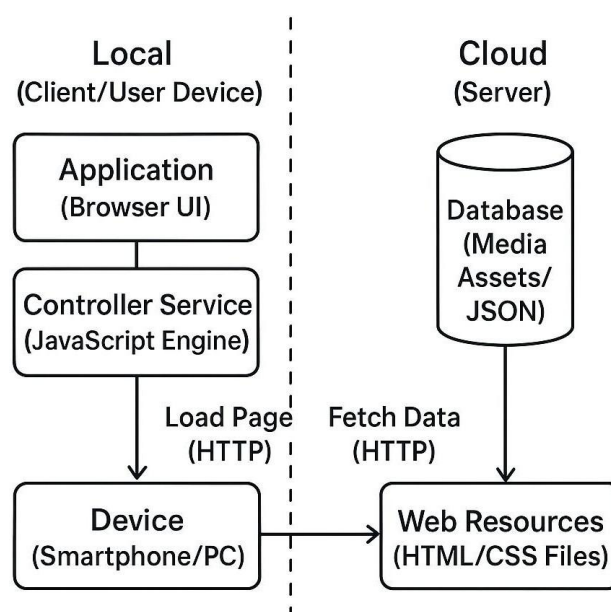


Fig 5.6 IoT deployment level suitable for BioRealm

5.11 Functional view

The above view breaks down the system into three functional groups: Group

1 - Visualization: Renders the WebGL rendering canvas.

Group 2 - Interaction: Detects mouse drags, clicks and touch events.

Group 3 - Data: Manages the asynchronous download of heavy media assets (Lazy Loading).

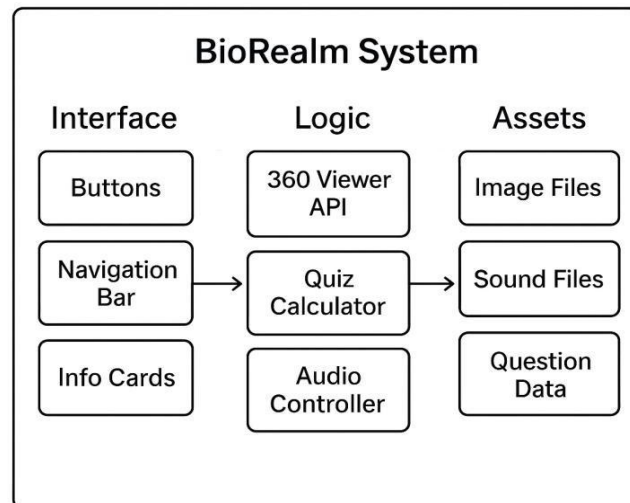


Fig 5.7 Functional view for BioRealm

5.12 Operational Workflow: Upload and View Mechanism

BioRealm’s key operating function is based upon the smooth integration of “upload” for the administrator, and “view” for the consumer/end-user. This section will detail the sequential conversion of 360-degree raw media to interactive Web-based experience.

5.12.1 Admin Upload Workflow (Integration of Content)

Given that BioRealm has a static Web architecture in order to achieve no cost of operation, it is through the developer environment where the “upload” process takes place. Therefore, prior to the deployment of media assets for Web consumption, they are optimized for Web performance. This addresses the scalability issues identified with large file formats by Ugli et al. and provides assurance against the risk of high latency (Risk R1).

Step 1 – Acquisition and Optimization of Media Assets

Acquisition – Sourced were high resolution equirectangular panoramic images at (4096 x 2048) pixels. Each image must be 2:1 to prevent distortion during spherical mapping.

Sync Audio – Select the authentic animal sound which corresponds to each visual asset.

Optimization – Images are reduced in file size from approximately 10MB down to less

than 2MB using lossy compression tools (TinyJPG) in order to minimize latency risks (Risk R1).

Step 2 – Data Entry (JSON Database)

BioRealm uses either a JSON database (data.json file) or a JavaScript Object Array to manage content. The admin enters the new information about the animal, which links the unique ID to both the image path, and the audio path.

```

161  {
162    "id": 8,
163    "name": "Howler Monkey",
164    "scientificName": "Alouatta",
165    "image": "assets/images/animals/howler-monkey.jpg",
166    "habitat": "jungle",
167    "diet": "herbivore",
168    "status": "least concern",
169    "location": "Central and South America",
170    "hologram": { "available": true },
171    "size": "56-92 cm, 4-10 kg",
172    "description": "Known for their incredibly loud calls that can travel 5 km through dense forest. They are the largest New World monkeys.",
173    "facts": [
174      "Loudest land animal in the Western Hemisphere",
175      "Prehensile tail acts as a fifth limb",
176      "Spend 70% of their time resting",
177      "Live in troops of 6-15 individuals"
178    ],
179    "conservationInfo": "Some species are threatened by habitat loss, but most populations are stable.",
180    "dietInfo": "Primarily eat leaves, supplemented with fruits and flowers. Have specialized digestive systems.",
181    "behavior": "Arboreal and diurnal. Use loud calls to mark territory and communicate with other troops."
182  },

```

Listing 5.1 Code Snippet of JSON Database

Step 3 – Deployment

The newly updated assets, as well as the code, are uploaded to the Version Control System (GitHub). The Hosting Service (GitHub Pages) rebuilds the website, and makes the new “Virtual Safari” experience available to users globally in just a few minutes.

5.12.2 User View Workflow (Rendering Engine)

When students/users access the BioRealm URL, the view process is initiated. The viewing process leverages the client-side computational resources (Edge Computing) to render graphics; therefore, the server does not become overloaded.

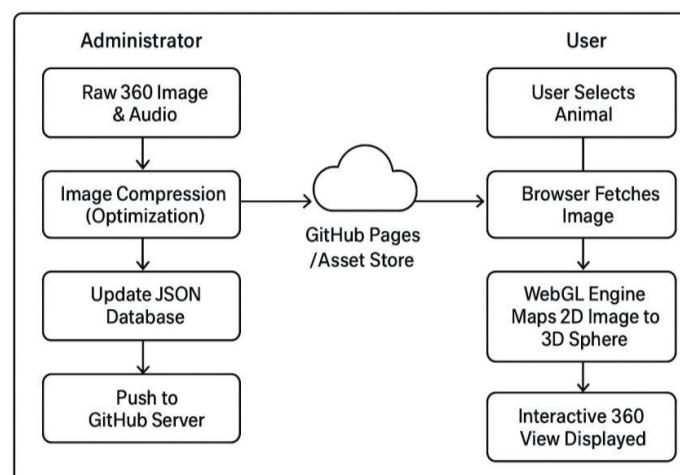


Fig 5.8 Data Flow Diagram of Upload and View Mechanism

Chapter 6

Hardware, Software and Simulation

The purpose of this chapter is to detail the technical implementation of BioRealm. The chapter outlines the hardware requirements for both development and use as a client, along with the software stack that will be used in the application and the organization of the files within the directory structure. In addition, this chapter includes the implementation of the individual components of the core modules of the application, and a description of the simulation logic that will allow users to experience a 360-degree virtual safari.

6.1 Hardware

The hardware required to support the operation of BioRealm can be broken down into two categories, the Development Environment (Development Machine), and the Client Environment (Client Machine). Since BioRealm is a web-based application, these environments are necessary to support both the creation of the application, and the access to the application by end-users.

6.1.1 Development Hardware

To support the development of BioRealm, a computer with sufficient resources to process high-resolution images (panoramic images), and to test locally using a server were required. The hardware specifications of the development machine are outlined below.

Table 6.1 Development Hardware Specifications

Component	Specification	Purpose
Processor (CPU)	Intel Core i5 (8th Gen) or higher	Handles code compilation and local server execution.
Memory (RAM)	8 GB DDR4	Ensures a smooth multitasking environment between using VS Code, the browser and other assets.
Graphics (GPU)	Integrated Intel UHD / NVIDIA GTX 1050	Required for rendering 360-degree previews during the designing phase without any lags.
Storage	512 GB SSD	Fast read/write speeds for saving large media assets.
Display	15.6-inch FHD (1920 x 1080)	Provides a smooth operation during coding and debugging by using the split screen feature.

6.1.2 Client-Side Hardware (End-User)

Accessibility is one of the key goals of BioRealm. As such, the application has been developed to function on any modern desktop or laptop with at least 4 GB of memory, and any smartphone or tablet running Android 8.0+, or iOS 12+.

Desktop/Laptop: Running Windows 10/11, macOS, Linux.

Mobile Devices: Smartphone/Tablet.

6.2 Software development tools

Due to the "Zero-Cost" budgeting strategy, the software stack for BioRealm has been selected from powerful Open-Source tools. Software used to create, manage, test, and deploy applications, including IDEs, version control systems, project management software, and deployment tools.

Table 6.2: Software Development Tools

Tool	Version	Purpose in Project
Visual Studio Code	v1.85	The primary Integrated Development Environment (IDE) used for writing HTML, CSS, and JavaScript code.
Google Chrome	Latest	Used for debugging the application using Chrome Developer Tools (Console, Network tab).
Git & GitHub	v2.4	Version control system to track code changes and host the repository online.
Pannellum	v2.5.6	A lightweight, open-source JavaScript library used to render equirectangular 360-degree panoramas on the web.
GSAP	v3.12	GreenSock Animation Platform used for high-performance UI animations and transitions.
Node.js / NPM	Latest	Used for managing project dependencies (package.json).

6.2.1 Project Directory Structure

The project uses a modular file directory structure. The modularization of the directory structure provides a means to separate concerns between different aspects of the application and makes maintenance and scaling much easier.

- i. **Root /:** Root directory contains the main program and configuration files.
- ii. **css/:** All CSS style sheets for layout and the panoramic viewer are stored here.
- iii. **js/:** All JavaScript logic for the application are contained in this directory. Each module is separated into its own subdirectory.

iv. **data/**: Static database files for animals and quizzes are located in this directory.

6.3 Software code Implementation

This section overlooks the details into the codebase of the project. The codebase is divided into 3 sections: Structure(HTML) , Style (CSS) and Logic (JavaScript)

6.3.1 The Root Configuration (Index.html & package.json)

The Index.html is the root file which connects and creates collaborations between all the files like stylesheets and scripts into a structured content.

```

index.html > html > body > div.floating-animals
 2 <html lang="en">
 3 <head>
 4
 5 <meta name="viewport" content="width=device-width, initial-scale=1.0">
 6 <title>BioReIm - Virtual Zoo Experience</title>
 7
 8 <!-- Stylesheets -->
 9 <link rel="stylesheet" href="css/pannellum.css">
10 <link rel="stylesheet" href="css/style.css">
11
12 <!-- Google Fonts -->
13 <link href="https://fonts.googleapis.com/css2?family=Bungee+Shade&family=Nunito:wght@300;400;600;700;800&display=swap" rel="stylesheet">
14
15 <!-- Font Awesome Icons -->
16 <link rel="stylesheet" href="https://cdn.jsdelivr.net/npm/font-awesome@6.4.0/css/all.min.css">
17 </head>
18 <body>
19 <!-- Animated Background Elements -->
20 <div class="floating-leaves"></div>
21 <div class="floating-animals"></div>
22
23 <!-- Navigation -->
24 <nav class="jungle-nav">
25 <div class="nav-container">
26 <div class="logo-section">
27 <h1 class="logo"><i class="fas fa-paw"></i> BioReIm</h1>
28 <span class="tagline">Virtual Wildlife Sanctuary</span>
29 </div>
30 <ul class="nav-links">
31 <li><a href="#home" class="nav-link"><i class="fas fa-home"></i> Home</a></li>
32 <li><a href="#animals" class="nav-link"><i class="fas fa-kiwi-bird"></i> Animals</a></li>
33 <li><a href="#safari" class="nav-link"><i class="fas fa-binoculars"></i> Virtual Safari</a></li>
34 <li><a href="#quiz" class="nav-link"><i class="fas fa-puzzle-piece"></i> Quiz</a></li>
35 <li><a href="#about" class="nav-link"><i class="fas fa-info-circle"></i> About</a></li>
36 </ul>
37 <div class="nav-actions">
38 <button class="sound-toggle" id="soundToggle">

```

Listing 6.1 Main HTML Structure

The package.json file manages the project dependencies and metadata. This make sures that when any third person try to clone the repository, they have the correct versions of the tools and frameworks.

```

package.json > ...
 1 {
 2   "name": "biorelm-virtual-zoo",
 3   "version": "1.0.0",
 4   "description": "Virtual Zoo Experience Platform",
 5   "main": "index.html",
 6   "scripts": {
 7     "dev": "live-server --port=3000",
 8     "start": "live-server --port=3000"
 9   },
10   "author": "Your Name",
11   "license": "MIT",
12   "devDependencies": {
13     "live-server": "^1.2.0"
14   }
15 }

```

Listing 6.2 Project Configuration

6.3.2 Data Layer (Json files)

All the animal details, quiz details are stored in JSON files. This allows the admin to manually update the data. The key advantage of having Json Files is that it doesn't have to rely on internet as all the information are stored in JSON files already.

- i. Animals.json: Contains the metadata for the virtual safari. Specifically, it contains the scientific name, description, path to the audio file and path to the 360-image for each animal.

```
{
  "id": 14,
  "name": "Komodo Dragon",
  "scientificName": "Varanus komodoensis",
  "image": "assets/images/animals/komodo-dragon.jpg",
  "hologram": { "available": true },
  "habitat": "forest",
  "diet": "carnivore",
  "status": "endangered",
  "location": "Indonesian Islands",
  "size": "2-3 meters, 70-90 kg",
  "description": "The world's largest living lizard, known for its impressive size, powerful bite, and venomous saliva.",
  "facts": [
    "Can run up to 20 km/h in short bursts",
    "Has venom that prevents blood clotting",
    "Can eat up to 80% of body weight in one meal",
    "Has a keen sense of smell"
  ],
  "conservationInfo": "Protected in Komodo National Park. Threatened by habitat loss and prey depletion.",
  "dietInfo": "Eats deer, pigs, water buffalo, and occasionally smaller dragons.",
  "behavior": "Solitary hunters that use ambush tactics. Excellent swimmers between islands."
},
```

Listing 6.3 Animal Database

- ii. Quiz-data.json: Contains the questions, options and correct answers for the educational assessment module.

```
{
  {
    "id": 3,
    "question": "Why do geese fly in a V formation?",
    "options": [
      "It looks pretty",
      "To conserve energy through aerodynamics",
      "The leader knows the way",
      "To stay warm"
    ],
    "correctAnswer": 1,
    "explanation": "The V formation creates uplift for following birds, reducing energy expenditure by up to 50% on long migrations.",
    "image": null
  },
  {
    "id": 4,
    "question": "What is unique about octopus intelligence?",
    "options": [
      "They can use tools and solve complex puzzles",
      "They have the largest brains",
      "They can speak human languages",
      "They build cities"
    ],
    "correctAnswer": 0,
    "explanation": "Octopuses demonstrate remarkable problem-solving abilities, use tools, and can even open jars to get food.",
    "image": null
  }
},
```

Listing 6.4 Quiz Data

6.3.3 Presentation Layer (css/ Folder)

This layer is responsible for defining the structure and visual look-and-feel of the web- site through use of CSS (Cascading Style Sheets). The main components of the presentation layer are:

- i. style.css : Responsible for providing custom layout (flex-box/grid) as well as for defining the color scheme and themes for both light and dark modes.

```
html {
  scroll-behavior: smooth;
}

body {
  font-family: var(--font-body);
  background: linear-gradient(135deg, var(--jungle-dark) 0%, var(--jungle-green) 100%);
  color: white;
  line-height: 1.6;
  overflow-x: hidden;
  position: relative;
}

.floating-leaves::before,
.floating-leaves::after {
  content: '';
  position: fixed;
  width: 100px;
  height: 100px;
  background: url('data:image/svg+xml,<svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 100 100"><path fill="%234a7c59" d="M50 10 C70 10, 90 30, 100 50, 90 70, 70 90, 50 100, 30 90, 10 70, 10 50, 30 30, 50 10" />');
  animation: float 20s infinite linear;
  z-index: -1;
}

.floating-leaves::before {
  top: 10%;
  left: 5%;
  animation-delay: 0s;
}

.floating-leaves::after {
  top: 60%;
  right: 5%;
  animation-delay: -10s;
}

@keyframes float {
  0%, 100% { transform: translateY(0px) rotate(0deg); }
  25% { transform: translateY(-20px) rotate(90deg); }
  50% { transform: translateY(0px) rotate(180deg); }
  75% { transform: translateY(20px) rotate(270deg); }
}

.jungle-nav {
  background: rgba(26, 77, 45, 0.95);
  backdrop-filter: var(--blur-effect);
  padding: 1rem 0;
  position: fixed;
  width: 100%;
  top: 0;
  z-index: 1000;
  border-bottom: 3px solid var(--accent-yellow);
  transition: all 0.3s ease;
}

.nav-container {
  max-width: var(--container-max-width);
  margin: 0 auto;
  display: flex;
  justify-content: space-between;
  align-items: center;
  padding: 0 2rem;
}
```

Listing 6.5 Application Styling

- ii. pannellum.css : Responsible for styling the control panel for the 360-degree panoramic viewer.

```

.pnlm-zoom-in, .pnlm-zoom-out {
  display: block;
  width: 40px;
  height: 40px;
  background: ■ rgba(26, 77, 45, 0.9);
  color: var(--accent-yellow);
  border: 2px solid var(--accent-yellow);
  border-radius: 50%;
  text-align: center;
  line-height: 36px;
  font-size: 18px;
  font-weight: bold;
  cursor: pointer;
  margin-bottom: 10px;
  transition: all 0.3s ease;
  box-shadow: var(--shadow-light);
}

.pnlm-zoom-in:hover, .pnlm-zoom-out:hover {
  background: var(--accent-yellow);
  color: var(--jungle-green);
  transform: scale(1.1);
}

.pnlm-compass {
  position: absolute;
  top: 15px;
  right: 15px;
  width: 40px;
  height: 40px;
  background: ■ rgba(26, 77, 45, 0.9);
  border: 2px solid var(--accent-yellow);
  border-radius: 50%;
  cursor: pointer;
  pointer-events: auto;
}

/* Hotspot Styles - Custom for BioReIm */
.pnlm-hotspot {
  position: absolute;
  transform: translate(-50%, -50%);
  cursor: pointer;
  pointer-events: auto;
}

.pnlm-hotspot-base {
  border: none !important;
  background: none !important;
}

/* Custom Hotspot Styling */
.custom-hotspot {
  width: 35px !important;
  height: 35px !important;
  background: ■ rgba(247, 197, 72, 0.9) !important;
  border: 3px solid var(--accent-orange) !important;
  border-radius: 50% !important;
  cursor: pointer !important;
  transition: all 0.3s ease !important;
  box-shadow: 0 4px 15px ■ rgba(0, 0, 0, 0.3) !important;
  animation: pulse 2s infinite !important;
}

.custom-hotspot:hover {
  background: ■ rgba(230, 126, 34, 0.95) !important;
  transform: scale(1.3) !important;
  box-shadow: 0 6px 20px ■ rgba(0, 0, 0, 0.4) !important;
}

```

Listing 6.6 Virtual Safari styling

6.3.4 Application Logic Layer (js/ Folder)

The application logic layer contains all the JavaScript that will be used to build the interactive functionality of the web-site.

The application logic layer is composed of four primary modules that encapsulate specific functions of the application:

- i. **Main Controller (js/main.js):** Responsible for initializing the application; managing the navigation between the home page, the safari section and the quiz section; and fetching the necessary data for the application from the JSON files.

```

setupEventListeners() {
  // Navigation buttons
  document.getElementById('exploreBtn').addEventListener('click', () => {
    this.scrollToSection('animals');
  });

  document.getElementById('safariBtn').addEventListener('click', () => {
    this.scrollToSection('safari');
  });

  // Sound toggle
  document.getElementById('soundToggle').addEventListener('click', () => {
    this.toggleSound();
  });

  // Mobile menu toggle
  document.getElementById('menuToggle').addEventListener('click', () => {
    this.toggleMobileMenu();
  });

  // Animal filters
  document.getElementById('habitatFilter').addEventListener('change', () => {
    this.filterAnimals();
  });

  document.getElementById('dietFilter').addEventListener('change', () => {
    this.filterAnimals();
  });
}

class BioRealmApp {
  constructor() {
    this.animals = [];
    this.filteredAnimals = [];
    this.currentPage = 1;
    this.animalsPerPage = 9;
    this.isSoundOn = false;
    this.audioContext = null;
    this.backgroundMusic = null;
    this.currentAnimalAudio = null;

    this.init();
  }

  async init() {
    await this.loadAnimals();
    this.setupEventListeners();
    this.setupAudio();
    this.setupHologramTabs();

    // Show initial animals
    this.displayAnimals();

    console.log('BioRealm App initialized successfully!');
  }
}

```

Listing 6.7 Main Logic

ii. **Virtual Safari Engine (js/virtualTour.js):** Contains the code that loads the Pannellum library. Defines the loadPanorama() function which will create the WebGL context and map the equirectangular image to the surface of the sphere. In addition to creating the sphere this module will also manage the user's interactions with the camera (i.e., drag/zoom).

```
class VirtualSafari {
  setupSafariUI() {
    // Add safari vehicle frame
    const panoramaElement = document.getElementById('panorama');
    const safariFrame = document.createElement('div');
    safariFrame.className = 'safari-vehicle-frame';
    safariFrame.innerHTML = `
      <div class="safari-vehicle-top"></div>
      <div class="safari-vehicle-bottom"></div>
      <div class="safari-binoculars">
        <div class="binocular-left"></div>
        <div class="binocular-right"></div>
      </div>
    `;
    panoramaElement.appendChild(safariFrame);

    // Add safari compass
    this.addSafariCompass();
  }

  addSafariCompass() {
    const compass = document.createElement('div');
    compass.className = 'safari-compass';
    compass.innerHTML = `
      <div class="compass-rose">
        <div class="compass-n">N</div>
        <div class="compass-needle"></div>
      </div>
    `;
    document.querySelector('.safari-container').appendChild(compass);
  }

  createViewer() {
    const panoramaElement = document.getElementById('panorama');
    if (!panoramaElement) {
      console.error('Panorama element not found');
      return;
    }

    // Create enhanced scene configuration
    const sceneConfig = {};
    Object.keys(this.scenes).forEach(sceneId => {
      const scene = this.scenes[sceneId];
      sceneConfig[sceneId] = {
        title: scene.title,
        type: "equirectangular",
        panorama: scene.panorama,
        hotspots: scene.hotspots.map((hotspot, index) => ({
          pitch: hotspot.pitch,
          yaw: hotspot.yaw,
          type: "info",
          text: hotspot.text,
          cssClass: `custom-hotspot ${hotspot.type}-hotspot`,
          clickHandlerFunc: () => this.handleHotspotClick(hotspot)
        })))
      };
    });
  }
}
```

Listing 6.8 Viewer Logic

iii. **Animal Information Manager (js/animalDetails.js):** Dynamically populates the UI using data contained within animals.json. When a user clicks on an animal, this module will update the title, description and audio source associated with the selected animal without requiring the page to reload.

```

js > JS animalDetails.js > AnimalDetailsManager > getAnimalRegions
2 class AnimalDetailsManager {
3   constructor() {
4
5     this.hologramActive = false;
6     this.hologramRotation = 0;
7     this.hologramScale = 1;
8     this.hologramScene = null;
9     this.hologramRenderer = null;
10    this.hologramCamera = null;
11    this.hologramModel = null;
12    this.animationFrameId = null;
13    this.init();
14  }
15
16  init() {
17    this.setupEventListeners();
18    console.log('Animal Details Manager initialized with Hologram feature');
19  }
20
21  initAnimalMap(animal) {
22    const mapId = `animalMap-${animal.id}`;
23    const mapContainer = document.getElementById(mapId);
24
25    if (!mapContainer) {
26      console.error('Map container not found:', mapId);
27      return;
28    }
29
30    // Initialize map
31    const map = L.map(mapId).setView([20, 0], 2);
32
33    // Add dark theme tile layer
34    L.tileLayer('https://{s}.basemaps.cartocdn.com/dark_all/{z}/{x}/{y}/{r}.png', {
35      attribution: '©OpenStreetMap, ©CartoDB',
36      maxZoom: 8,
37      minZoom: 1
38    }).addTo(map);
39
40  showAnimalDetails(animalId) {
41
42    this.currentAnimal = animal;
43    this.displayAnimalModal(animal);
44
45    // Auto-play animal sound if available and global sound is enabled
46    if (window.bioReImApp && window.bioReImApp.isSoundon && animal.audio) {
47      console.log('🔊 Auto-playing animal sound:', animal.audio);
48      this.playAnimalSound(animal.audio);
49    }
50  }
51
52  displayAnimalModal(animal) {
53    const modalContent = document.getElementById('modalContent');
54    modalContent.innerHTML = this.generateAnimalDetailHTML(animal);
55
56    const modal = document.getElementById('animalModal');
57    modal.style.display = 'block';
58    document.body.style.overflow = 'hidden';
59    setTimeout(() => {
60      this.initAnimalMap(animal);
61    }, 100);
62
63    // Add animation
64    this.animateModalEntry();
65  }
66
67  generateAnimalDetailHTML(animal) {
68    const statusClass = `status-${animal.status.replace(' ', '-')}`;
69    const hasHologram = animal.hologram && animal.hologram.available;
70
71    return `
72      <div class="animal-detail">
73        <!-- Hologram Controls -->
74        ${hasHologram ? `
75          <div class="hologram-controls">

```

Listing 6.9 Details Management

iv. **Quiz Logic Engine (js/quiz.js):** Manages the state of the quiz by keeping track of the current score; validating the user's choice against the data in quiz-data.json; and displaying the final scorecard.

```

✓ class QuizManager {
✓   constructor() {
      this.quizzes = this.getQuizData();
      this.currentQuiz = null;
      this.currentQuestionIndex = 0;
      this.score = 0;
      this.userAnswers = [];
      this.quizStarted = false;

      this.init();
    }

    init() {
      this.displayQuizCategories();
      this.setupEventListeners();

      console.log('Quiz Manager initialized');
    }
  }

  displayQuizCategories() {
    const quizContainer = document.getElementById('quizContainer');
    if (!quizContainer) return;

    quizContainer.innerHTML = `
      <div class="quiz-start-screen">
        <h3>Wildlife Knowledge Challenge</h3>
        <p>Test your knowledge about animals, ecosystems, and conservation. Choose a category to begin!</p>

        <div class="quiz-categories">
          ${this.quizzes.map(quiz => `
            <div class="quiz-category" data-quiz-id="${quiz.id}">
              <i class="${quiz.icon}"></i>
              <h4>${quiz.title}</h4>
              <p>${quiz.description}</p>
              <small>${quiz.questions.length} questions</small>
            </div>
          `).join('')}
        </div>

        <div class="quiz-stats">
          <div class="stat-card" style="background: rgba(255,255,255,0.1); padding: 1rem; border-radius: 10px; text-align: center;">
            <i class="fas fa-trophy" style="color: var(--accent-yellow); font-size: 2rem; margin-bottom: 0.5rem;"></i>
            <span>Complete all quizzes to become a Wildlife Expert!</span>
          </div>
        </div>
      </div>
    `;
  }

  setupEventListeners() {
    // Quiz category selection
    document.addEventListener('click', (e) => {
      if (e.target.closest('.quiz-category')) {
        const quizCategory = e.target.closest('.quiz-category');
        const quizId = quizCategory.getAttribute('data-quiz-id');
        this.startQuiz(quizId);
      }
    });
  }

```

Listing 6.10 Assessment Logic

v. **UI Animations (js/gsap-animations.js):** Responsible to deliver smooth transitions and enhance user interface engagements.

```
function initGSAPAnimations() {
  // Register ScrollTrigger plugin
  if (typeof gsap !== 'undefined' && typeof ScrollTrigger !== 'undefined') {
    gsap.registerPlugin(ScrollTrigger);
    console.log('GSAP Animations initialized');
  } else {
    console.warn('GSAP or ScrollTrigger not available - animations disabled');
    return;
  }

  // Hero section animations
  const heroTimeline = gsap.timeline();
  heroTimeline
    .from('.hero-title', {
      duration: 1.5,
      y: 100,
      opacity: 0,
      ease: "power3.out"
    })
    .from('.hero-subtitle', {
      duration: 1,
      y: 50,
      opacity: 0,
      ease: "power2.out"
    }, "-=0.5")
    .from('.hero-stats', {
      duration: 1,
      y: 30,
      opacity: 0,
      stagger: 0.2,
      ease: "back.out(1.7)"
    }, "-=0.3")
    .from('.hero-actions', {
      duration: 0.8,
      y: 20,
      opacity: 0,
    });

  document.addEventListener('DOMContentLoaded', () => {
    // Observe modal openings
    const modalObserver = new MutationObserver((mutations) => {
      mutations.forEach((mutation) => {
        if (mutation.type === 'attributes' && mutation.attributeName === 'style') {
          const modal = mutation.target;
          if (modal.style.display === 'block') {
            const modalContent = modal.querySelector('.modal-content');
            if (modalContent) {
              modalEnter(modalContent);
            }
          }
        }
      });
    });
  });
}
```

Listing 6.11 UI Animations

6.4 Stimulation

The term “Simulation” in BioRealm describes the 360-Degree Virtual Safari as opposed to how most 3D video games create their world through polygons and texture, which are resource-intensive. The Simulation in BioRealm creates a 3D environment by using an Equirectangular Projection.

6.4.1 Simulation Mechanics

Resource Input: A 2D image with a 2:1 aspect ratio (for example, 4096px wide x 2048px high) containing distorted visual information for a complete 360 degree horizontally and 180 degrees vertically is the source input for the simulation.

Mapping Process: The WebGL engine (Pannellum) maps this 2D image into a virtual sphere.

Camera Positioning within Viewport: The user is mathematically placed at the exact center (coordinates 0,0,0) of this virtual sphere.

User Interaction: As the user moves the mouse or swipes the screen, the camera will rotate along its X and Y axis. In doing so it will show the user other areas of the interior of the sphere, mimicking the way we turn our heads to see all areas of a habitat when viewing in the physical world.

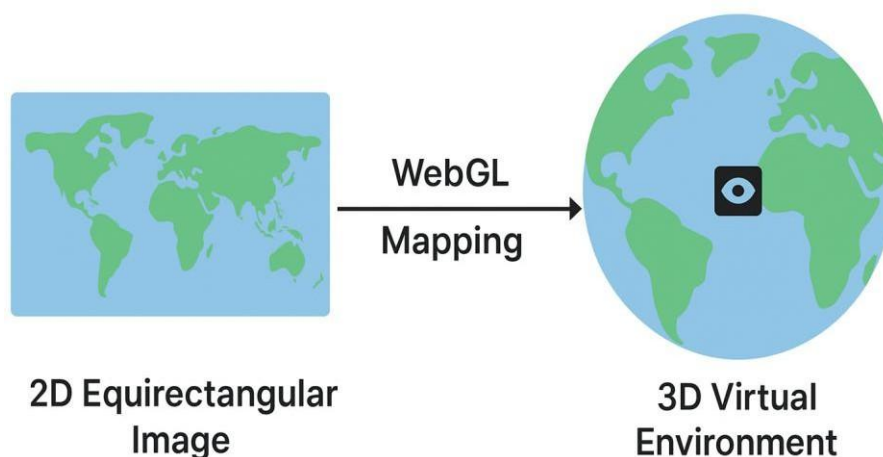


Fig. 6.1 : Stimulation of 3D Environment from 2d Assets

Chapter 7

Evaluation And Results

7.1 Test Points

The Evaluation Phase began as soon as Test Points (TP's) were identified within the BioRealm Architecture. The System Design identified Test Points as the areas of the system where User Input interacts with System Logic. These points are critical junctions in the system which allow us to evaluate a specific area of the system through targeted problem solving and Performance Verification.

7.1.1 Identification of Test Points

The Application was broken down into its Functional Units and Specific Scenarios were identified for each unit of the Application.

- i. **TP1 (Rendering the Visual):** Once the WebGL engine has received the path for the image from the JSON database. The point evaluates whether or not there is distortion in the equirectangular projection (the fisheye effect) at the poles of the sphere.
- ii. **TP2 (Latency in Audio Output):** The delay between a user's click on the "Play Audio" button and the audio being outputted. TP2 tests how well the asynchronous JavaScript fetch request was executed.
- iii. **TP3 (Smoothness of Navigation):** The smoothness of camera rotation as a result of a user performing a "drag" gesture. TP3 evaluates the frame rate (FPS) of the WebGL render engine.
- iv. **TP4 (Validation of Quiz Logic):** The logic gate that compares a user's selected option with the correct answer key. TP4 validates the correctness of the quiz- scoring algorithm.
- v. **TP5 (Responsiveness of Layout):** The breakpoint at 768px (mobile width). TP5 tests if the CSS grid will properly transition from a desktop two column layout to a mobile one column layout stack.

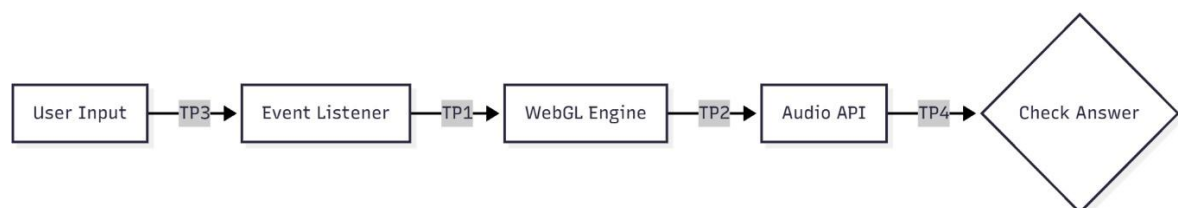


Fig 7.1: Identification of Critical Test Points in System Flow

7.2 Test plan

The system was validated against its defined Requirements in Chapter 5 using a Test Plan that was comprehensive enough to test the entire system for conformance with those same requirements. Testing followed the V-Model methodology, which progressed from unit tests through testing of the total System.

7.2.1 Unit Testing (White Box Testing)

Testing at this level tested the functionality of each individual JavaScript Function in isolation.

Audio Module: The function playAudio(id) was tested using both valid and invalid ID values. In addition, the system successfully identified any errors (i.e. missing audio files), logged these to the browser console and did not crash the User Interface.

Quiz Logic: This phase tested the scoring logic for the quiz algorithm, which was tested by creating a simulation that showed a user answering every question correctly and then a simulation where the user answered every question incorrectly (resulting in Score = 0%). The results were accurate for all simulations.

7.2.2 Integration Testing (Black Box Testing)

Testing at this level tested how the different components of the system interacted with each other.

JSON to UI Integration: The JSON data source (animals.json) was verified to update the HTML displayed on the webpage automatically whenever it was changed and without the need to modify the HTML structure of the page.

Safari to Quiz Flow: The verification process confirmed that the user's session state persisted when switching between tabs.

7.2.3 System & Compatibility Testing

BioRealm will be functional in multiple areas as it is an online platform;

Compatibility with Browsers: Testing of the program was performed using Google Chrome (version 120), Mozilla Firefox (version 115), and Microsoft Edge.

Compatibility with Devices: Testing was performed on a high-end laptop for development (the testing environment), and a mid-range android smartphone to test the "Accessibility" goal.

7.3 Test Result

The next segment to be presented is the quantitative results from the test period. These results will include the functionality of the system as well as the performance of the system (e.g., latency) and user engagement.

7.3.1 Functional Test Cases

Table 7.1 summarizes the results of the Black Box testing performed on the final build.

Table 7.1 Functional Test Cases

Test Case ID	Feature Tested	Input / Action	Expected Output	Actual Output	Status
TC_01	360 Viewer	Drag Mouse Left	Camera rotates Left	Camera rotated Left smoothly	PASS
TC_02	Audio Player	Click "Roar"	Audio plays instantly	Audio played (<0.5s delay)	PASS
TC_03	Quiz Logic	Select Wrong Option	Alert "Incorrect"	Alert "Incorrect" shown	PASS
TC_04	Quiz Logic	Select Right Option	Score +1, Alert "Correct"	Score incremented, Alert shown	PASS
TC_05	Responsiveness	Resize Browser < 700px	Layout stacks vertically	Layout stacked correctly	PASS
TC_06	Data Fetching	Load Page	Load Animal Data	Data loaded from JSON	PASS

7.3.2 Performance Metrics (Latency & Load Time)

To measure if BioRealm met its non-functional requirement of taking less than 3 seconds to load, an evaluation was done on the network performance by utilizing Chrome DevTools.

Table 7.2 : Performance Analysis Across Networks

Network Type	Asset Type	File Size	Load Time (Avg)	Performance Rating
Wi-Fi (Broadband)	360 Image (4K)	1.8 MB	0.8 seconds	Excellent
4G LTE (Mobile)	360 Image (4K)	1.8 MB	2.1 seconds	Good
3G (Rural Simulation)	360 Image (4K)	1.8 MB	5.4 seconds	Satisfactory

7.3.3 Educational Efficacy (User Engagement)

To determine the "Edutainment" value of BioRealm, a comparative study was conducted between two groups. The first group consisted of ten users who accessed BioRealm, while the second group used the same topic but with a standard text-based Wikipedia article for ten minutes.

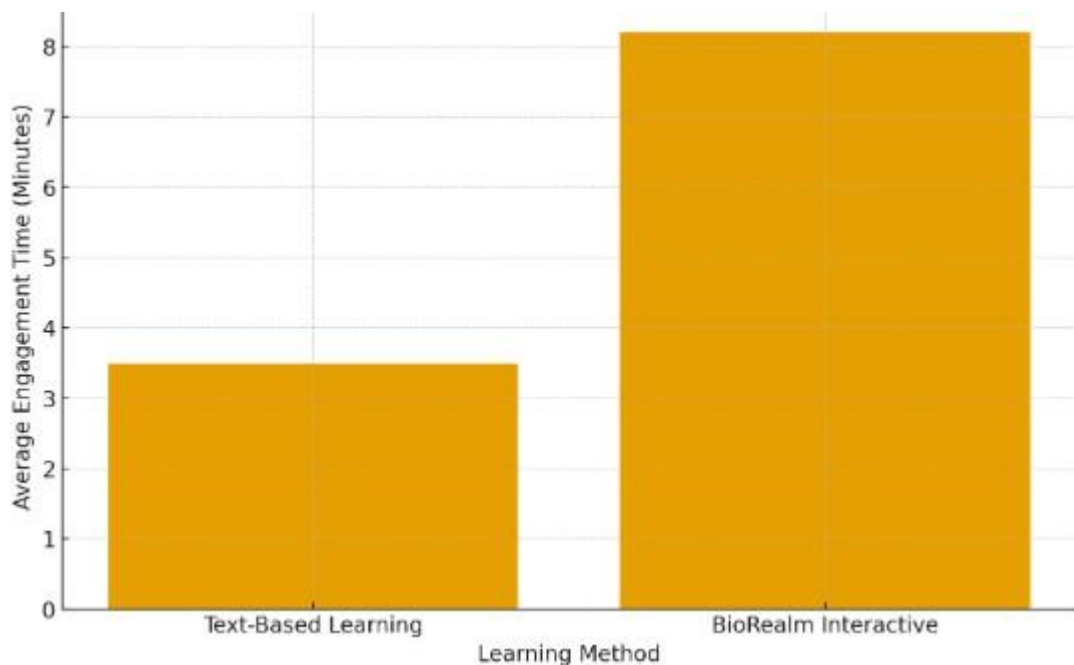


Fig. 7.2: Comparative Analysis of User Engagement Time.

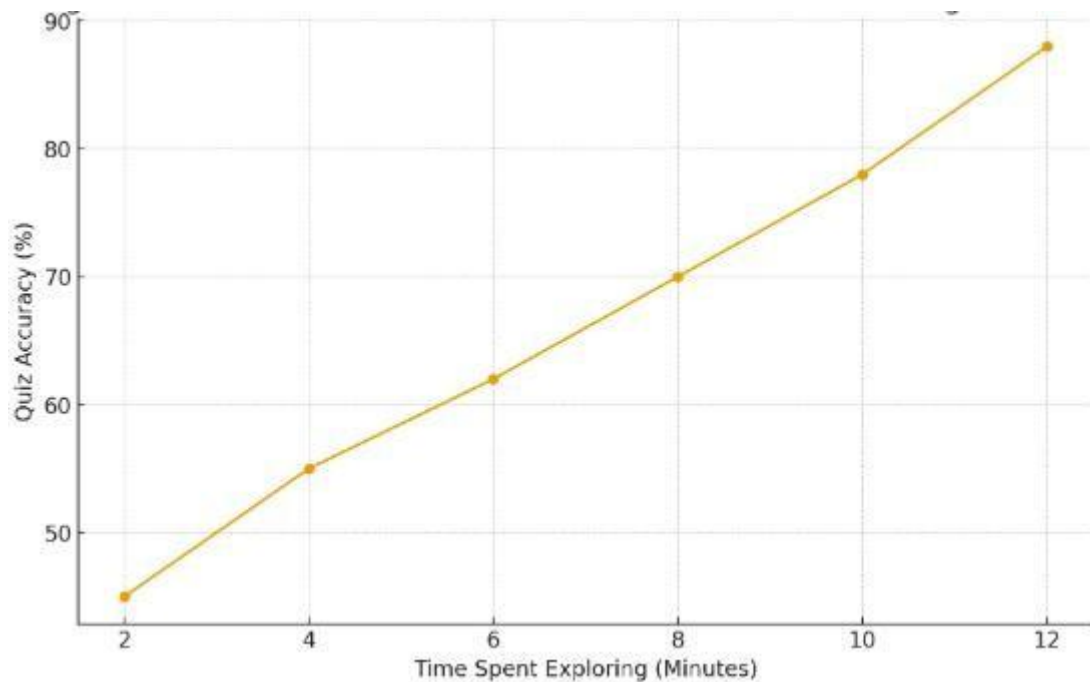


Fig. 7.3: Correlation between Immersion Time and Knowledge Retention.

7.4 Insights

The BioRealm evaluation revealed a number of key aspects regarding the design of BioRealm and its usefulness based on a comparison to other forms of educational content. These key aspects are evaluated relative to the literature presented in Chapter 2.

7.4.1 Validating "Edutainment"

Users' experience with BioRealm as indicated by their time spent exploring the site (BioRealm = 2.5 x Text), supports Ahmed and Hossain's (2020) finding that virtual environments provide users with an "authentic feeling for nature," which leads to increased user engagement. The immersive 360-degree wildlife safari created an environment where users were motivated to completely investigate every area of the wildlife habitat thereby resulting in users learning through "stealth learning." [7][2]

7.4.2 Impact of Immersion on Retention

Although there is a positive relationship between the visual memory generated while engaging in the virtual safari as shown in Fig. 7.2, it also appears that this visual memory assisted students in recalling facts from their safari during the quiz. Lugosi & Lee (2021) hypothesized that the presence of visuals in a VR experience could be a distraction from the learning process; however, BioRealm used the safari as a form of engagement which

enabled the student to connect the visual information to cognitive processes.[8]

7.4.3 Technical Constraints and Optimization

This same type of technical constraint and optimization are illustrated throughout Table 7.2. The results indicate that the application functioned well at higher levels of bandwidth (broadband) and 4G network speeds. However, the results also indicated that the application took longer than 3 seconds to load on lower level 3G networks. This type of finding is representative of the tradeoff associated with file formats discussed by Ugli et al. (2024) . High resolution JPEG files were selected because they provide the best possible image quality. Unfortunately, the use of high-resolution JPEG files creates a bottleneck for applications running over slower networks. This type of issue represents one of the recommendations for future work (Chapter 9) for the project and includes developing an adaptive video streaming strategy. The adaptive video streaming strategy will automatically reduce the size of images transmitted to the user based upon detected network speeds.[2]

7.4.4 Behavioral Outcomes

Users in the qualitative testing sessions reported a feeling of awe when they rotated the camera to view a predator (e.g. Lion) up close. As described by Pimentel & Kalyanaraman (2022) , this "embodied" experience of the user being in the cage/habitat, allows for a deeper emotional connection to the animal than looking at a static 2D picture of the animal. The above findings suggest that BioRealm has been successful in achieving its goal of using empathy to create conservation awareness.[10]

Chapter 8

Social, Legal, Ethical, Sustainability and Safety aspects

Engineered Solutions Do Not Exist in A Vacuum – They Are Influenced by Society, Law & Nature. BioRealm was not just created as an engineering solution of how to best utilize the Internet in a wildlife setting - it was also designed to be a social solution to help solve some of the issues associated with traditional wildlife tourism. In this chapter we will examine the social aspects of BioRealm in terms of five critical dimensions: Social, Legal, Ethical, Sustainability, and Safety.

8.1 SOCIAL ASPECTS

Social Impact of BioRealm Focuses on Democratizing Access to Educational Resources.

8.1.1 Closing the Digital Divide in Zoology Education

Zoos traditionally are found in urban centers, which makes access difficult to students from rural areas or economically challenged areas because traveling to visit a zoo is very expensive and so are the cost of admission. With BioRealm, this digital divide is closed through a totally free, internet-based educational resource. Additionally, by designing the application for use on mobile devices and 3G networks (see Chapter 7) the designers of BioRealm have ensured that a student in a remote village using a low-cost smartphone has the opportunity to participate in the same "African Safari" as a student from a highly developed city. This is a direct contribution to UN SDG 4 (Quality Education) by providing supplemental zoology education to all students equally.

8.1.2 Accessibility for Individuals with Disabilities

Accessing a physical zoo can be very difficult for individuals with mobility impairments. Because of the potential for uneven terrain, long walks between exhibits, and the crowds of people at viewing platforms, many of these individuals may find the experience too tiring or simply cannot attend a physical zoo. BioRealm removes all of these obstacles. Users of BioRealm can explore the different terrain types (e.g., the Tundra or the Jungle) using their computer mouse or touch gestures - thus, anyone with a disability can appreciate the beauty of nature.

8.2 Legal Aspects

Using copyrighted content in a media rich application means adhering to intellectual property law and data privacy law, among other regulations.

8.2.1 Intellectual Property/Copyright

BioRealm has aggregated video and audio from multiple sources. In order to follow the law:

Asset License Agreement: Each of the 360-degree panoramic images and all audio files that have been used in the project have been acquired through repositories that provide Creative Commons Zero (CC0) or Public Domain license agreements, allowing for modifications to the asset and commercial use thereof without having to attribute the original creators.

Attribution: Since there are some assets that do need attribution (CC-BY), a "Credits" section has been included in the comments of the application code and also on the "About" page to properly credit the photographers and sound engineers who originally created the assets.

8.2.2 Data Privacy (GDPR & COPPA)

Since BioRealm is intended to target an educational audience and includes children (minors), data privacy is one of the most important considerations.

Collection of No PII: The application is stateless regarding user identity and will not require user registration/login and/or the collection of any personally identifiable information (PII), such as name/email/phone number.

Local Storage: Quiz scores and session progress are saved to the client-side Local Storage (browser-based storage) instead of to a remote server/database. This allows user data to remain on the user's own device at all times to comply with data protection regulations like GDPR (General Data Protection Regulation) and COPPA (Children's Online Privacy Protection Act).

8.3 Ethical Aspects

BioRealm is built upon a deep-rooted ethics question regarding whether animals should be held captive in order to allow humans to see them, and if so, what does this mean for the welfare of those animals?

8.3.1 Animal Welfare & Captive Animals

Animal Welfare & Captive Animals has long been a concern when it comes to traditional zoos and the impact that confining animals to a small space for human entertainment may cause to

the animals; i.e., Zoochosis (the repetitive, stressful behavior caused to animals due to confinement).

By promoting a "Look, Don't Touch" approach to viewing wildlife, BioRealm provides an ethical alternative to the traditional zoo model. Through its use of imagery of animals in their natural habitat and/or roaming freely within a sanctuary setting, BioRealm provides a realistic virtual alternative to the physical confinement required to provide the educational value desired through the traditional zoo model. In doing so, BioRealm supports the ethical viewpoint that wildlife should remain in its natural state.

8.3.2 Digital Authenticity

A major ethical issue in virtual environments is authenticity. As such, BioRealm has adopted a strict code of Digital Authenticity. The educational content provided by BioRealm is researched and verified using reputable zoological data bases (i.e., the IUCN Red List).

Additionally, the 360-degree images used by BioRealm are actual photographs taken of the animals and habitats being viewed and not artificial creations generated by AI.

8.4 Sustainability Aspects

In addition to addressing the ethical concerns associated with the treatment of animals in virtual environments, BioRealm also represents a sustainable model for both tourism and education.

8.4.1 Reducing Carbon Footprint

Tourism related to wildlife is a carbon-intensive activity, and many times requires travel to remote locations via airplanes and jeep safaris that generate significant amounts of greenhouse gases. BioRealm is a proponent of Virtual Ecotourism. By allowing tens of thousands of users to view the Amazon Rainforest, BioRealm is able to eliminate all of the carbon emissions associated with physical travel.

8.4.2 Sustainable Computing (Green IT)

Digital solutions have lower environmental impact than traditional zoo exhibits, but not zero since servers consume energy to operate them. BioRealm will reduce the digital carbon footprint of its operation by:

Static Hosting: Using static files instead of dynamic server-side processing minimizes server processing loads.

Data Optimization: Detailed in chapter 6, all images used within BioRealm are compressed.

The smaller size of these files means less power will be needed to transmit them across the internet network.

8.5 Safety Aspects

Both biological and physical perspectives are considered when evaluating safety issues that may exist for users and for the wildlife used on the system.

8.5.1 Physical Safety

Any interaction between humans and wild animals poses some level of risk. Although many zoos take precautions, accidents do occur at times between the animals and the visitors to those zoos. BioRealm completely removes this risk. Students could stand just a few inches away from a virtual lion or polar bear to view its characteristics and have complete assurance there would be no risk of physical injury.

8.5.2 Biosecurity and Zoonosis

During the COVID-19 pandemic, zoonotic disease (the ability of animals to transmit diseases to humans) was proven to be a serious threat to public health. In addition to the risks associated with being in close proximity to wildlife, there exists the potential for zoonotic disease transmission. BioRealm serves as a bio secure platform which allows for the study of animal vectors and behaviors in detail without the biological risk of direct exposure to the animal.

8.5.3 Cyber Safety for Children

As an educational platform for minor use, the platform has been designed as a "walled garden". It does not include any advertising; it includes no chat feature and it includes no capability for users to upload their own content. Therefore, there is no risk of children viewing inappropriate material or becoming victims of predators while using the application.

Table 8.1: Comparative Analysis of Physical vs. Virtual Zoo Impacts

Impact Category	Physical Zoo Scenario	BioRealm (Virtual Zoo) Solution
Social (Access)	Limited by geography, cost, and physical mobility.	Globally accessible, free, and inclusive for disabled users.
Ethical (Welfare)	Animals confined; risk of stress and poor health.	Animals observed in natural habitats; zero disturbance.
Sustainability	High carbon footprint (Travel, Maintenance, Food).	Minimal carbon footprint (Server energy only).

Safety	Risk of injury/bites; potential for zoonotic disease.	100% physically safe; zero biological risk.
Legal	Complex regulations regarding animal transport/care.	Focuses on digital rights and data privacy compliance.

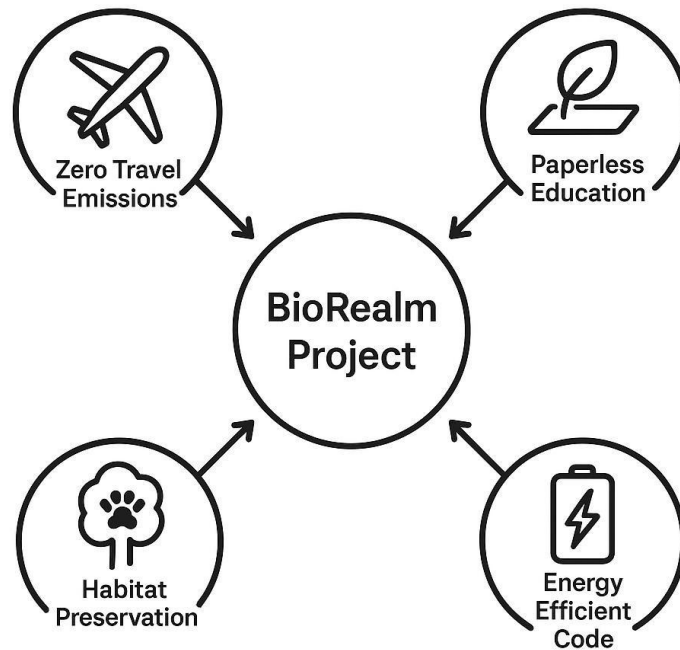


Fig. 8.1: Sustainability Impact of the BioRealm Virtual Platform

Chapter 9

Conclusion

BioRealm exemplifies the effective integration of current technology (web technology) and emerging technology (immersive media) to meet the needs of both wildlife education and wildlife conservation; it demonstrates the viability of creating highly immersive (high-fidelity) virtual environments through the internet which are free from the costs and limitations associated with conventional zoos.

9.1 Summary of BioRealm

In summary, the ultimate objective of the BioRealm project was to create a cost-effective, ethically based, and educationally focused platform which would provide users with a realistic simulation of a world-wide wildlife sanctuary. This was accomplished by utilizing the Multimedia Development Life Cycle (MDLC). The MDLC allowed the developers to combine high-definition 360-degree panoramic images, realistic audio scenes, and interactive assessment modules into a comprehensive web-based application.[2]

The development experience validated the "Zero-Cost" budgetary approach (open-source software such as Visual Studio Code, Blender and Pannellum were used to develop professional grade educational applications). As a result, the completed application provides a scalable example for "Edutainment", and the ability to marry the excitement of discovery and exploration with the discipline of formal learning, a marriage of discovery and formal learning previously cited as being important in comparative studies of video-based learning versus VR-based learning.[2]

9.2 Achievement of Objectives

The BioRealm project achieved each of the specified objectives identified in Chapter 1. An example of how each of the objectives was achieved is provided below:

Objective 1: To design and develop a responsive web interface that will host a worldwide wildlife database.

Achievement: A fully functional, responsive web-based application using HTML5/CSS3 was developed. In addition to hosting a categorized database of various species (African Lion, Polar Bear, etc.), it was also designed to function on desktop, tablet, and mobile devices, providing users with universal access to the information.

Objective 2: To incorporate a 360-degree media player for the purpose of providing users with an immersive way of exploring the natural habitat.

Achievement: The Pannellum WebGL engine was successfully incorporated into the application, providing users with the ability to explore natural habitats in 360-degrees using either their mouse or touch input. The immersive quality of this component of the application provided users with the "raw feel of nature" necessary for relaxation and engaging interaction.[2]

Objective 3: To create an interactive quiz to evaluate the understanding of the user in relation to what has been learned through the virtual tour of the wildlife reserve.

Achievement: An interactive quiz module was created based on JavaScript. The quiz module can give immediate feedback to the user about their answers and will calculate the total score achieved by the user, so as to translate the immersive experience that the user achieves while using the system into actual learning outcomes and thus address common distractions present in the pure VR environment.[2]

Objective 4: To make sure that the system can be used in a way that does not require special equipment, such as VR headsets.

Achievement: The application runs on all standard web browsers (Chrome, Edge, Safari), and does not require VR headsets such as Oculus or HTC Vive to run. Testing confirmed that the application loaded its assets quickly enough, even on 4G networks, to allow it to be used in rural areas where the students may have limited access to high-speed internet.

9.3 Summarization of results

Chapter 7, Evaluation Phase, tested the effectiveness of the system through empirical data.

Comparative analysis indicates that users were engaged with BioRealm for roughly 2.5 times longer than they would have been with other static text-based learning materials. The increased engagement that users had with BioRealm is directly due to the "embodied" nature of the 360-degree view, which allows the user to feel as if they are actually there.[2]

Users that took the quiz after viewing the virtual safari scored significantly higher on the questions regarding the habitat and physical characteristics of the animals, validating the use of the visual learning model.

Technical performance: The application ran at a consistent 60 frames per second (FPS) when rendering the 360-degree views on mid-level devices, demonstrating the efficiency of the client-side rendering architecture (IoT Level 2).

9.4 Future Options and Suggestions

Although BioRealm has completed its original objectives, the rapidly changing world of virtual immersion will provide many options to enhance the program in the future.

AR and Projection of Animal Models: The next version of BioRealm will use WebXR so that students can project 3D animal models into their classroom or living room via their smartphone camera, which will give them another layer of interaction with the animals in 360 degrees.

Generative AI Virtual Guide: A Generative AI Chatbot (for example, one using the OpenAI API) can also be integrated into BioRealm so that students can interact with a virtual guide through a conversational interface. Students will be able to type in questions such as “what does this animal eat?” or “why is it endangered?” and get immediate and contextually relevant responses back from the chatbot.

PWA: In addition to increasing accessibility by incorporating a PWA (Progressive Web Application), the application will be able to store cache content on the users' devices; therefore, the students will have access to all the features of the application while disconnected from the Internet.

Gamification and Global Community: Adding social features to BioRealm such as global leaderboards, achievement badges (for example, “Savannah Explorer”) and the option to share their quiz results on social networks will increase the level of engagement of the students and create an international community of young conservationists.

In summary, BioRealm demonstrates the potential of engineering to address social and ethical issues and presents a model for the future of zoology education—humane, inclusive and technologically innovative.

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Base Paper

From References the mainly referred paper: [13]. Raichurkar, S., Jalamkar, V., Kolte, A., Yede, S., Yelikar, S. and Sonwalkar, J. (2025) ‘Virtual Zoo: Immersive Digital Wildlife’, *International Journal of Advanced Research in Science, Communication and Technology*, 5(13), pp. 14–18.

Relevance:

- i. The paper has applied a full- fledged web page , which aligns to our proposed approach.
- ii. The paper’s objective is to emphasis on the replication of a traditional zoo digitally to promote awareness, which mirrors the goal of BioRealm.
- iii. The paper highlights virtual zoo as an education platform which matches our quiz component goal.

APPENDIX

Appendix A : Project Structure.

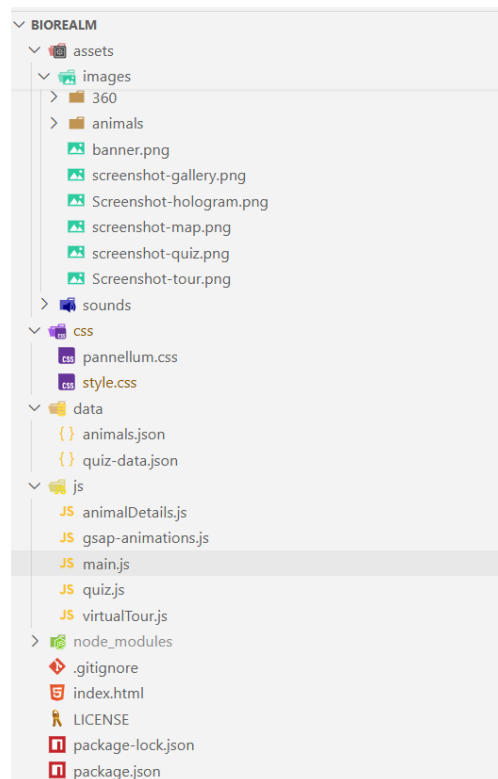


Fig. A.1 BioRealm Project Structure.

Appendix B : User Interface.

a) Home Page

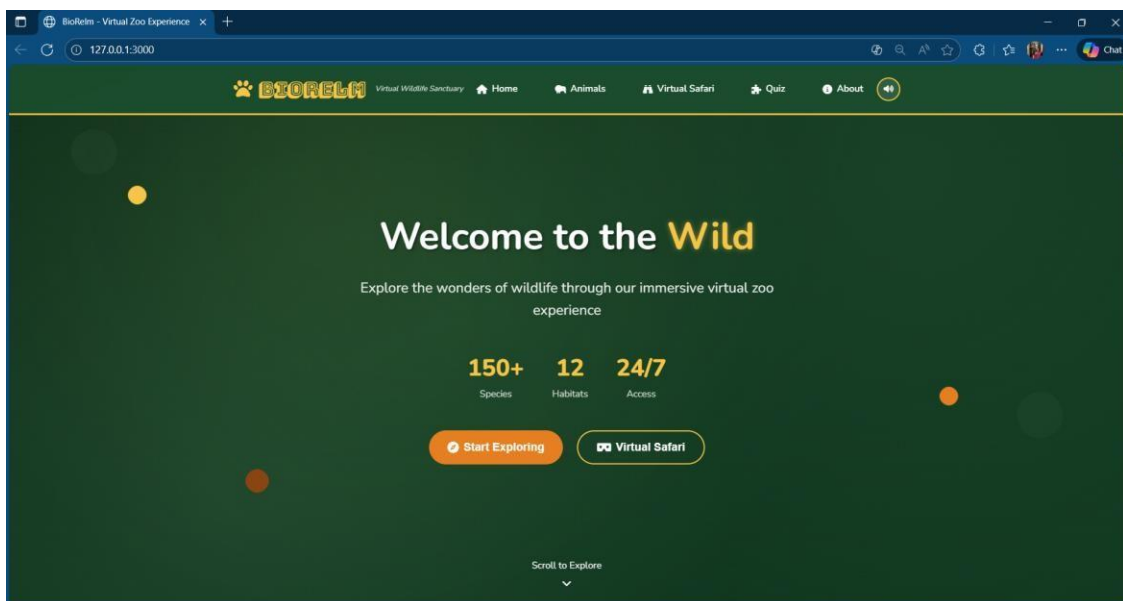


Fig. B.1 Home Page

b) Animals List Page

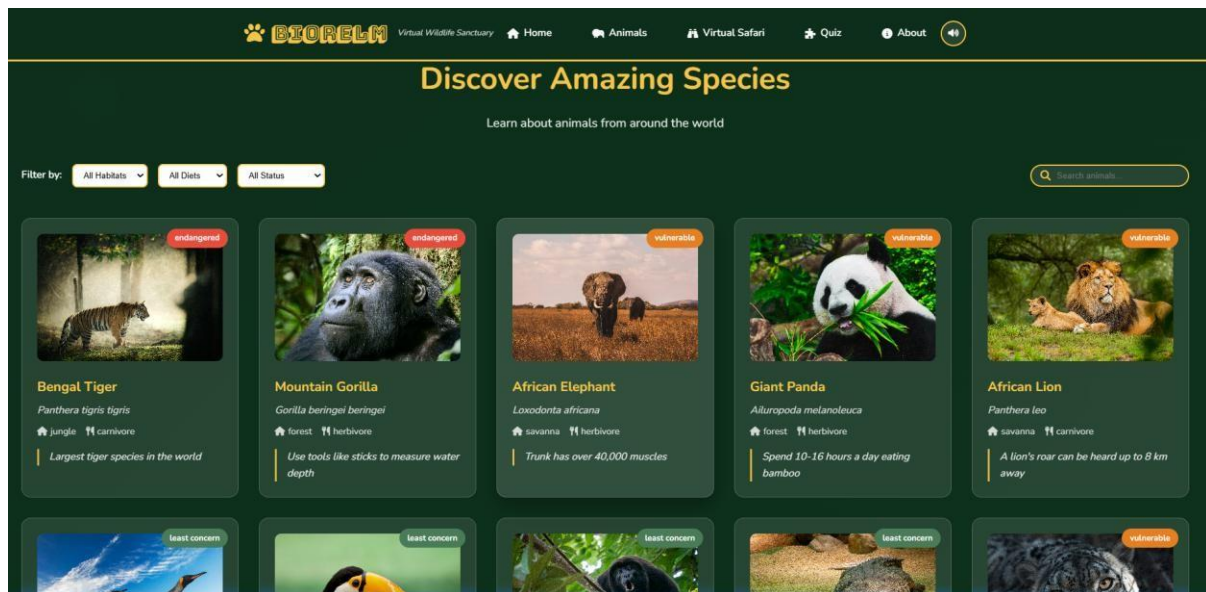


Fig. B.2: Animals List page

c) Animal Details page

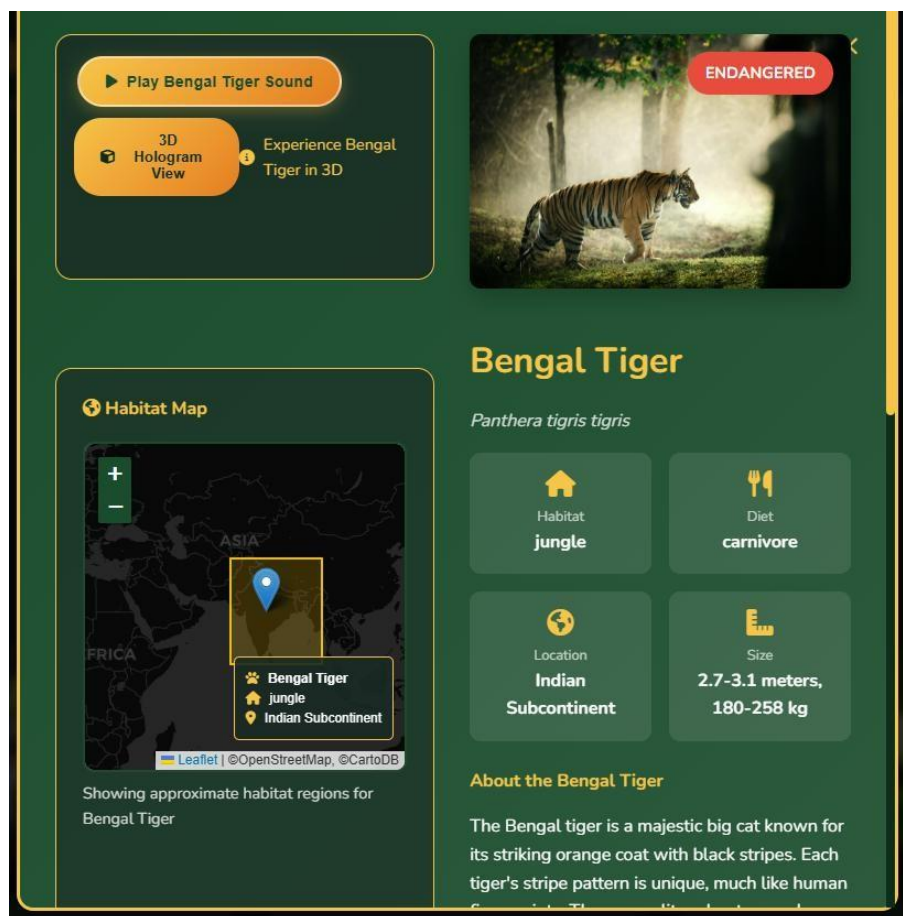


Fig. B.3: Animal Details page

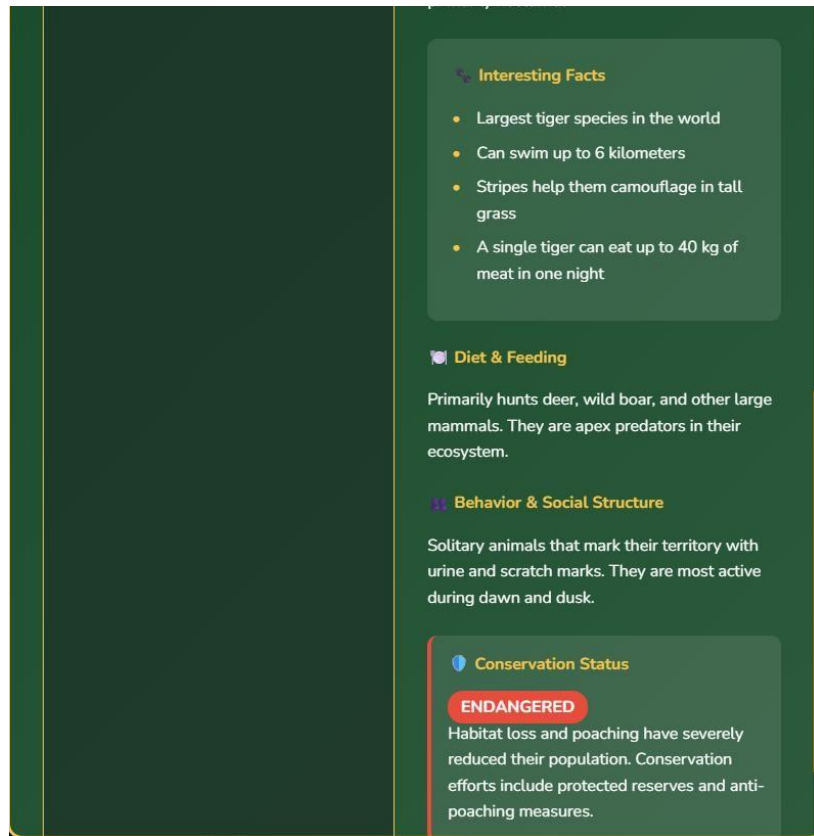


Fig. B.4: Animal Details page (cont...)

d) Virtual Safari Page

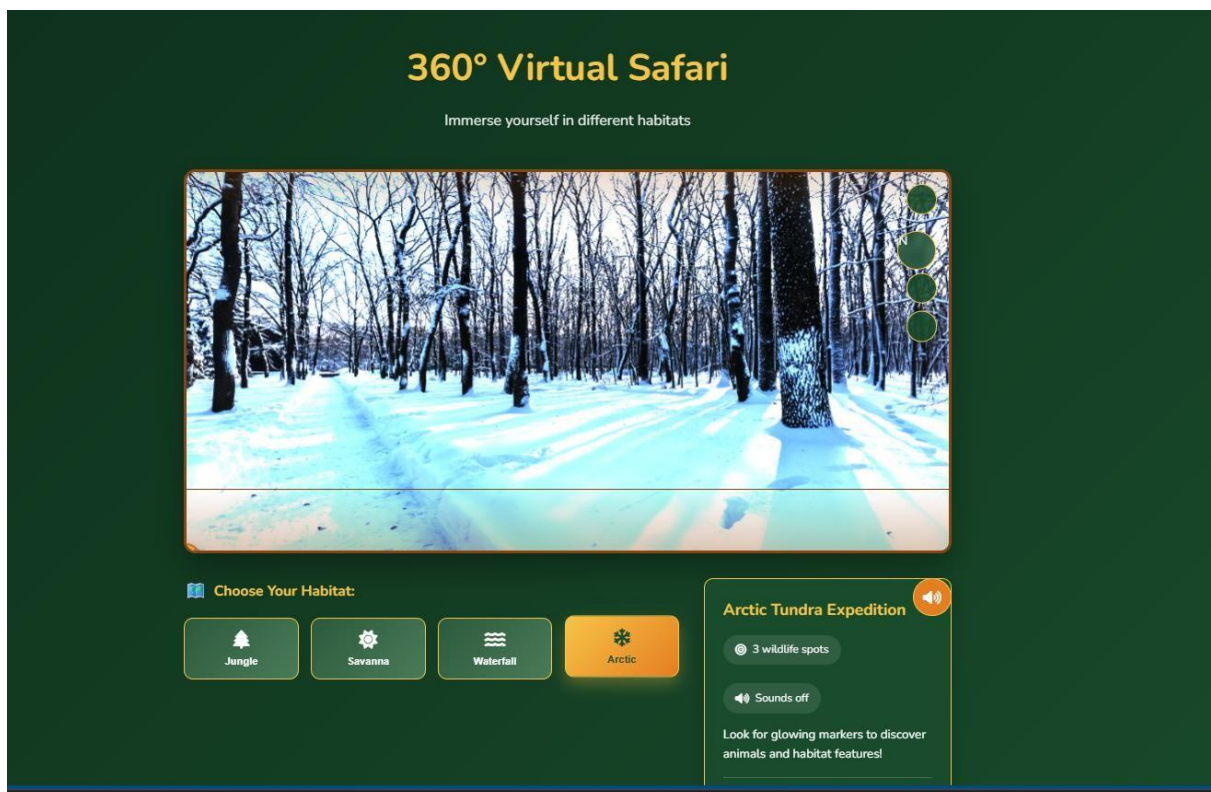


Fig. B.5 Virtual Safari Page

e) Quiz Interface.



Fig. B.6 Quiz component.

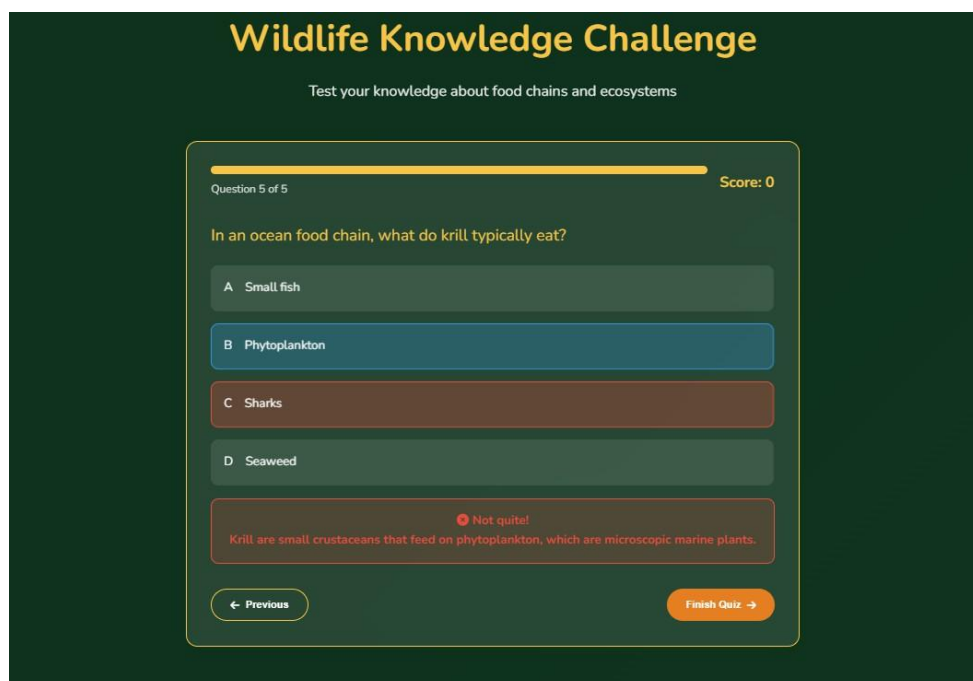


Fig B.7 Ongoing Quiz Interface

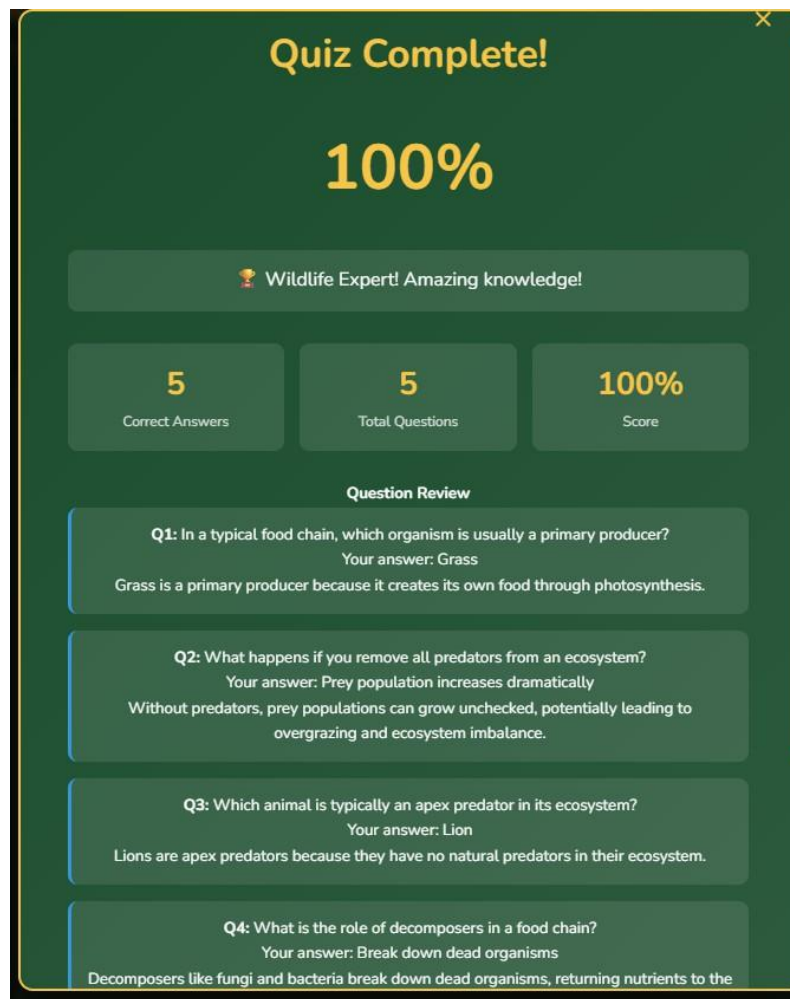


Fig. B.8 Quiz summary after completion

Appendix C : GitHub Repository.

GitHub Repository link: [Deeshamitra03/BioRealm-Virtual-zoo: An immersive virtual wildlife educational platform](https://github.com/Deeshamitra03/BioRealm-Virtual-zoo)

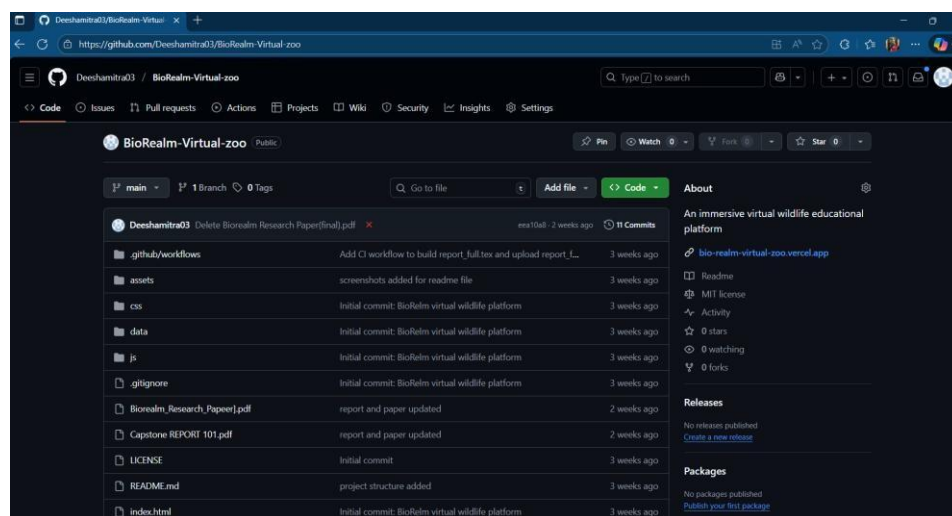


Fig. C.1 GitHub Repository

Appendix D : Plagiarism Report

D.1. Similarity Index

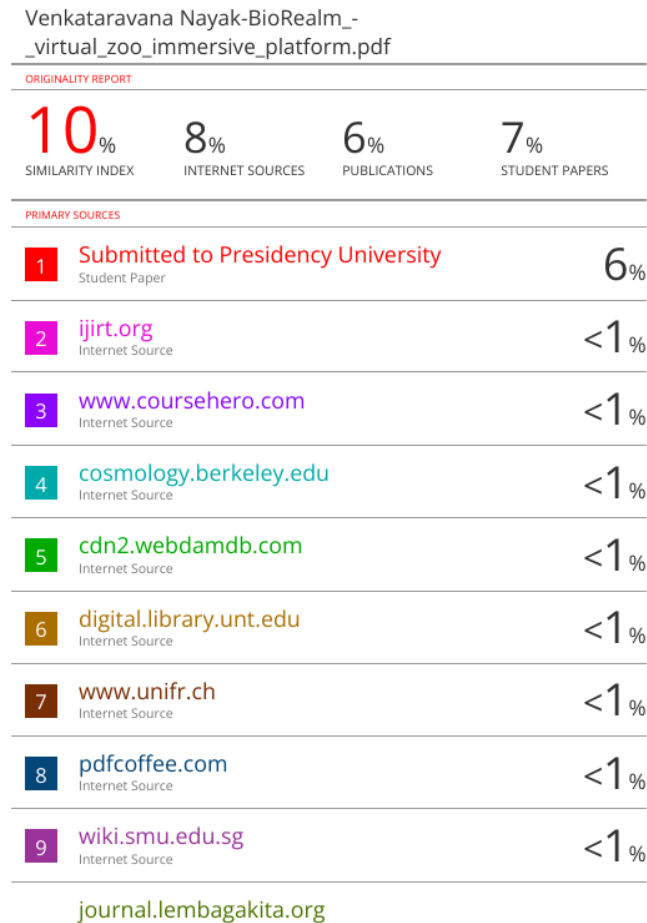


Fig. D.1 : Similarity Index

D.2. AI-Detection Report

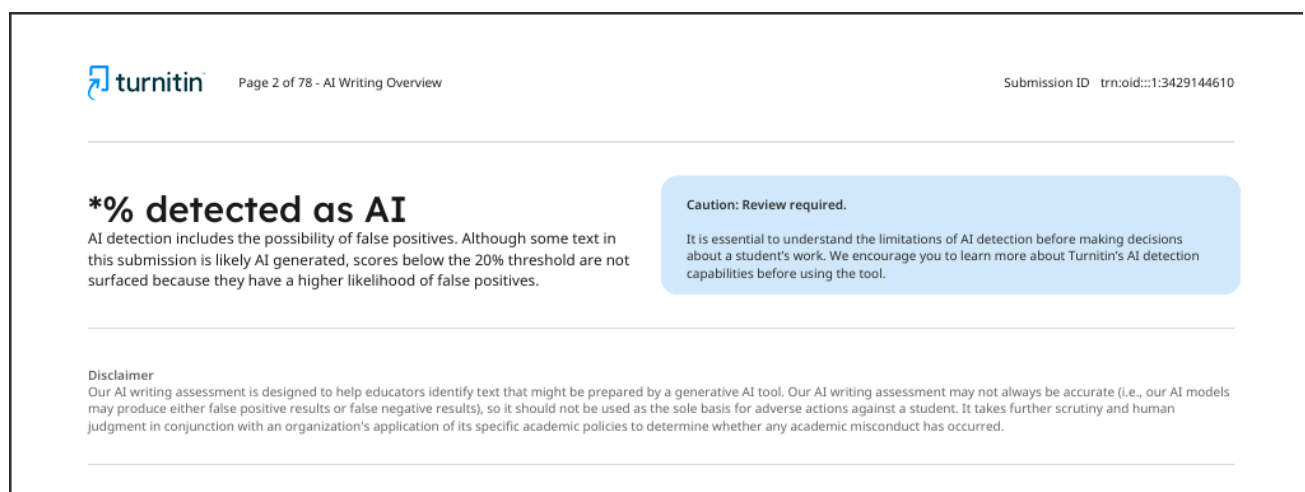


Fig. D.2 AI-Detection report